

CARL-WSN: Context-Aware Reinforcement Learning for Hybrid Routing in Heterogeneous IoT-Driven Wireless Sensor Networks

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Abstract

Wireless Sensor Networks (WSNs) deployed in IoT-driven smart city environments carry heterogeneous application traffic with fundamentally different quality-of-service requirements. Existing routing protocols — including recent reinforcement learning (RL) based approaches — optimise a single objective such as energy efficiency or network lifetime without adapting routing behaviour to the urgency class of individual packets. This paper proposes CARL-WSN, a Context-Aware Reinforcement Learning protocol that uses a tabular Q-Learning agent to dynamically select the routing paradigm — proactive, reactive, or hybrid — on a per-packet basis. The agent observes a three-dimensional context vector comprising application urgency class, normalised residual energy, and link stability score, and learns a class-dependent multi-objective reward function that weights latency, overhead, and energy preservation according to traffic priority. A safety constraint forces Class A emergency packets onto proactive routing regardless of the agent’s policy, guaranteeing sub-100 ms delivery at all times. CARL-WSN further integrates energy-aware relay selection and bounded critical-node packet deferral to extend network lifetime. The compact Q-table (216 bytes of RAM, stored in single precision) enables deployment on resource-constrained sensor hardware. CARL-WSN is evaluated in MATLAB against seven baseline protocols —

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AODV, DSDV, ZRP, two domain-specific baselines (EH-Routing, MSLBA), and two recent RL-based approaches (RLCR and FQ-UCR) — under four traffic scenarios across 20 independent seeds (640 runs), with all protocols re-implemented on a common distance-dependent lossy-channel model so that delivery, latency, and energy emerge from identical physical assumptions. Under mixed traffic, CARL-WSN extends network lifetime beyond the classical baselines (first node death at round 418, the highest of all protocols; half-node death at round 2198, second only to MSLBA), delivers the lowest Class A emergency latency among the reactive and learning-based protocols (24.2 ms, a 25.6% reduction versus AODV), and incurs routing overhead far below the flooding- and table-based protocols (0.53 control packets per data packet, versus 14.82 for AODV and 2.31 for DSDV). These gains are obtained by trading a modest amount of packet delivery: CARL-WSN achieves 90.2% PDR under its tuned deferral policy, within roughly five points of the baselines, with energy consumption statistically comparable across all protocols and throughput comparable to the classical and energy-harvesting baselines. The learned Q-table converges within approximately 1150 rounds, and the agent autonomously discovers context-dependent policies that adapt routing overhead to traffic-class composition — a capability absent in all seven baselines. A sensitivity analysis confirms that CARL-WSN’s performance is robust to its principal design parameters.

Keywords: Wireless sensor networks, Reinforcement learning, Q-Learning, Context-aware routing, Hybrid routing protocol, IoT, Smart city, Energy-aware relay selection, Heterogeneous traffic

1. Introduction

The proliferation of Internet of Things (IoT) devices in smart city infrastructure has transformed Wireless Sensor Networks (WSNs) from single-application monitoring systems into multi-service platforms that simultaneously carry traffic with fundamentally different quality-of-service (QoS) requirements [1, 2]. A single WSN deployment may generate emergency alerts requiring sub-100 ms delivery, periodic telemetry tolerating delays of tens of seconds, and event-driven uploads with best-effort requirements. This traffic heterogeneity demands routing protocols that adapt their behaviour to the urgency class of individual packets — a capability absent from both classical and recent learning-based routing approaches.

Routing protocols for WSNs fall into three paradigms, each with inherent trade-offs. Proactive protocols such as DSDV [3] maintain routing tables continuously, delivering low latency at the cost of high energy overhead from periodic broadcasts. Reactive protocols such as AODV [4] and DSR [5] discover routes on demand, conserving energy in low-traffic conditions but introducing route-discovery delays of 20–100 ms that violate emergency latency constraints [6]. Hybrid protocols such as ZRP [7] partition the network into zones and apply different modes within and across boundaries, but use a fixed zone radius that does not adapt to node energy state or traffic urgency.

Recent work has applied reinforcement learning (RL) to WSN routing with promising results. Daanoun and Baghdad [8] proposed RLCR, a Q-Learning-based clustering and routing protocol that optimises cluster head selection and inter-cluster forwarding for energy efficiency. Wang and Duan [9] proposed FQ-UCR, combining fuzzy logic for unequal cluster formation with Q-Learning for inter-cluster routing. While both protocols demonstrate the potential of RL for adaptive WSN routing, they share a fundamental limitation: routing decisions are made based on network-level metrics (residual energy, distance to base station) without any awareness of the application-level urgency class of the packet being routed. An emergency alert and a routine temperature reading receive identical routing treatment — a systematic inefficiency that leads to either wasted energy on low-urgency traffic or dangerous delays on safety-critical packets.

A 2025 comprehensive survey by Khedr et al. [10] explicitly identified dynamic application-context-based hybrid mode switching as an open research direction in IoT-driven WSNs, noting that reactive and hybrid context-aware approaches have received significantly less attention than proactive ones. No published protocol uses reinforcement learning to select between fundamentally different routing paradigms — proactive, reactive, and hybrid — on a per-packet basis according to application urgency. This paper addresses that gap directly.

The main contributions of this paper are as follows:

- A formal Markov Decision Process (MDP) formulation for routing paradigm selection in heterogeneous IoT-WSNs, with an $18\text{-state} \times 3\text{-action}$ state-action space derived from a three-dimensional context vector comprising application urgency class U , normalised residual energy E_r , and link stability score L_s .
- CARL-WSN, a Q-Learning-based routing protocol that learns the op-

timal mapping from context states to routing paradigms (proactive, reactive, or hybrid) through online interaction with the network environment, using a class-dependent multi-objective reward function that weights latency, overhead, and energy preservation according to traffic priority. A safety constraint forces Class A emergency packets to proactive routing, guaranteeing sub-100 ms delivery regardless of the agent’s learned policy.

- An energy-aware relay selection mechanism combined with critical-node packet deferral that extends network lifetime by preferring high-residual-energy next-hop nodes and deferring non-urgent packets when the transmitting node is near energy depletion — a context-aware lifetime extension capability that only urgency-aware protocols can implement.
- A comprehensive evaluation against seven baselines — three classical protocols (AODV, DSDV, ZRP), two domain-specific baselines (EH-Routing, MSLBA), and two recent RL-based protocols (RLCR, FQ-UCR) — across four traffic scenarios and 20 independent seeds (640 runs). Crucially, all eight protocols are re-implemented on a common distance-dependent lossy-channel model with per-hop acknowledgement and retransmission, so that packet delivery, latency, and energy consumption emerge from identical physical assumptions rather than from protocol-specific idealisations. Under this unified methodology, CARL-WSN extends network lifetime beyond all baselines (highest first-node-death round) and attains the lowest emergency-traffic latency among the reactive and learning-based protocols, while incurring routing overhead far below the flooding- and table-based protocols. These gains are obtained by trading a modest, quantified reduction in packet delivery, which we characterise explicitly.

The remainder of this paper is organised as follows. Section 2 reviews related work on hybrid, context-aware, and RL-based routing in WSNs. Section 3 presents the system model. Section 4 describes the CARL-WSN protocol design, including the MDP formulation and Q-Learning mechanism. Section 5 presents the simulation setup. Section 6 presents and analyses the results. Section 7 discusses key findings, trade-offs, and limitations. Section 8 concludes the paper.

2. Related Work

2.1. Proactive and Reactive Routing Protocols

Routing protocols for WSNs have been extensively surveyed and classified [11]. Proactive protocols maintain routing tables continuously through periodic broadcasts. DSDV [3] established the foundational destination-sequenced distance-vector approach, guaranteeing loop-free routes at all times. OLSR [12] extended this paradigm through multipoint relay selection, substantially reducing flooding overhead in dense networks. While proactive protocols deliver consistently low latency by eliminating route discovery delays, their continuous maintenance overhead leads to significant energy waste on low-traffic nodes [11].

Reactive protocols discover routes exclusively on demand. AODV [4] pioneered this approach by initiating route requests only when data must be transmitted, conserving energy in sparse or low-traffic conditions. However, the route-discovery delay introduced by on-demand RREQ/RREP flooding can make reactive protocols less suitable for the most latency-sensitive traffic unless routes are cached and reused [6]. A comparative analysis by Nurul Hidayah et al. [13] confirmed that neither proactive nor reactive protocols alone achieve optimal performance across varying network conditions, motivating the need for adaptive hybrid approaches. Recent work [14] demonstrated that AODV, DSDV, and DSR all exhibit significant performance degradation under heterogeneous traffic loads, highlighting the fundamental limitation of single-mode routing for IoT-driven smart city deployments.

2.2. Hybrid Routing Protocols

Hybrid protocols attempt to combine the latency advantages of proactive routing with the energy efficiency of reactive routing. ZRP [7] pioneered zone-based hybrid routing by maintaining proactive routes within a fixed-radius zone and using reactive inter-zone routing for destinations beyond the zone boundary. While ZRP reduces control overhead compared to purely proactive protocols, its fixed zone radius is configured statically and does not adapt to node energy state or traffic urgency.

Subsequent hybrid approaches incorporated additional optimisation dimensions. Chaurasiya et al. [15] proposed a hybrid clustering technique for multilevel heterogeneous IoT WSNs, combining cluster-based organisation with multi-hop routing to reduce energy consumption. Lei [16] proposed a hybrid energy-aware approach combining particle swarm optimisation with fuzzy

119 clustering for IoT routing, demonstrating 52% energy reduction compared
 120 to DEEC, though at the cost of requiring offline training data. Nadanam
 121 et al. [17] recently proposed a quantum-optimised hybrid routing framework
 122 achieving 30% energy reduction, though the computational complexity of
 123 quantum optimisation makes it unsuitable for resource-constrained sensor
 124 hardware.

125 A common limitation across all these hybrid approaches is that mode
 126 selection or routing decisions are made based on a single optimisation objective
 127 — typically energy or load — without awareness of the application-level urgency
 128 of the traffic being routed.

129 *2.3. Context-Aware Routing in IoT-WSNs*

130 Context-aware computing for IoT was formally defined by Perera et al. [18],
 131 who identified five contextual dimensions — identity, location, activity, time,
 132 and relations — as fundamental to intelligent IoT decision making. This
 133 foundational work motivates the use of contextual information in routing
 134 decisions, though the specific application to RL-based WSN hybrid routing
 135 mode selection had not been addressed until the present work.

136 Context-aware routing has since been applied across several IoT and WSN
 137 domains. Monteiro et al. [19] proposed context-aware network selection in
 138 heterogeneous wireless networks, demonstrating that context-driven decisions
 139 outperform static policies, but their work focused on network selection rather
 140 than intra-WSN routing mode switching. Sithik and Kumar [20] proposed
 141 a multi-context aware parent selection scheme for the RPL protocol using
 142 grey set theory, incorporating multiple contextual parameters into routing
 143 metric computation. However, their approach targets RPL parent selection
 144 exclusively and does not implement dynamic switching between fundamentally
 145 different routing paradigms based on application urgency class. Karagiannis
 146 et al. [21] proposed context-aware routing for fog computing systems, reducing
 147 latency by 23% and hop count by 73%, though their work operates at the fog
 148 layer rather than the WSN sensor layer and does not model heterogeneous
 149 traffic classes with conflicting QoS requirements. Hasan et al. [22] proposed
 150 a context-aware trust prediction framework for opportunistic IoT routing
 151 using Bayesian inference, achieving 88.9% delivery ratio, though the trust-
 152 based context dimension differs fundamentally from the urgency-energy-
 153 stability context vector proposed in CARL-WSN. Akpakwu et al. [23] proposed
 154 a context-aware congestion control scheme for CoAP/UDP IoT networks,
 155 applying context sensing to transport-layer decisions rather than routing.

Smart city IoT routing has also received significant attention. Karunkuzhali et al. [24, 25] proposed QoS-aware routing frameworks for IoT-WSNs in smart cities, demonstrating that multi-class service differentiation improves network performance. However, none of these works implement dynamic routing paradigm switching based on per-packet application urgency. Elfouly et al. [26] proposed environment-aware real-time routing for multi-sink WSNs in smart cities, incorporating deadline constraints and link quality metrics. This work shares the multi-sink topology with our prior work [27] and considers real-time delivery requirements. Smart city IoT routing has also been addressed by Hanif et al. [28] through opportunistic IoT-WSN integration, Inga et al. [29] through WSN sizing and routing optimisation, and Ekler et al. [30] through energy-aware IoT routing algorithms. However, none of these works implement dynamic routing paradigm switching based on per-packet application urgency, nor do they model heterogeneous application urgency classes or implement context-driven mode switching.

2.4. Reinforcement Learning for WSN Routing

The application of reinforcement learning to WSN routing has emerged as one of the most active research frontiers in recent years [31, 32, 33]. RL-based protocols offer the ability to learn routing policies through online interaction with the network environment, adapting to changing conditions without requiring explicit models or predefined rules.

Daanoune and Baghdad [8] proposed RLCR, a reinforcement learning-based clustering and routing protocol for heterogeneous WSNs. RLCR uses Q-Learning in two phases: first for cluster head (CH) selection based on residual energy, distance to the base station, and node density; and second for inter-cluster multi-hop routing where each CH learns the optimal next-hop CH toward the base station. RLCR demonstrated significant lifetime improvements over LEACH and DEEC variants. However, RLCR does not differentiate traffic by application urgency class — all packets receive identical routing treatment regardless of whether they carry emergency alerts or routine telemetry. Furthermore, RLCR’s RL agent operates exclusively within the clustering paradigm and does not consider switching between fundamentally different routing modes.

Wang and Duan [9] proposed FQ-UCR, a hybrid approach combining fuzzy logic (FL) for unequal cluster head selection with Q-Learning for inter-cluster routing. FQ-UCR addresses the hotspot problem by forming smaller clusters near the base station and larger clusters at greater distances, using a fuzzy

Table 1: Comparison of CARL-WSN with Related Routing Works

Protocol	AI/ML method	Context dims	Mode switching	Traffic classes	Urgency aware	Energy-aware relay	Paradigm switching
RLCR [8]	Q-Learning	3	CH selection	1	No	No	No
FQ-UCR [9]	Fuzzy + Q-Learning	3	CH selection	1	No	No	No
Sithik & Kumar [20]	Grey set theory	2	RPL parent	1	No	No	No
Karagiannis et al. [21]	None	2	Fog forwarding	1	No	No	No
Hasan et al. [22]	Bayesian	1 (trust)	Route selection	1	No	No	No
ZRP [7]	None	1 (zone)	Hybrid fixed	1	No	No	No
CARL-WSN (Ours)	Q-Learning	3	Per-packet RL	3	Yes	Yes	Yes

inference system to compute CH selection probability. The Q-Learning agent then selects the optimal forwarding CH for inter-cluster data relay. FQ-UCR demonstrated improved energy efficiency and lifetime compared to EEUC and CHEF. Similar to RLCR, however, FQ-UCR does not model heterogeneous traffic classes or adapt routing behaviour to application urgency.

Sudha et al. [34] combined Q-Learning with machine learning for cluster head selection and routing in 5G-enabled WSNs, demonstrating improved throughput over LEACH, but without considering per-packet application urgency as a routing criterion. Saadallah and Alabady [35] identified energy-efficient context-sensitive routing as an open challenge in their comprehensive IoT routing study, while Kumar et al. [36] confirmed through bibliometric analysis that application-context awareness remains an emerging research gap in energy-efficient IoT-WSN routing.

Table 1 summarises the key distinguishing features of CARL-WSN compared to the most closely related works. CARL-WSN is the only protocol that combines RL-based learning with per-packet application urgency awareness for full routing paradigm switching.

2.5. Research Gaps

The preceding review reveals five specific gaps that motivate CARL-WSN. First, Khedr et al. [10] conducted the first comprehensive survey of context-aware routing in IoT-driven WSNs and explicitly identified dynamic application-context-based mode switching as an open research direction, noting that reactive and hybrid context-aware methods have received significantly less attention than proactive approaches. Second, existing RL-based WSN routing protocols (RLCR [8], FQ-UCR [9]) apply Q-Learning within a single routing paradigm (clustering) without awareness of application urgency class — the RL agent cannot distinguish between emergency and routine traffic. Third,

220 Chu et al. [37], Khanna et al. [38], and Prashanth et al. [39] all noted that routing
 221 adaptation in heterogeneous WSNs remains dominated by single-objective
 222 optimisation approaches that cannot simultaneously satisfy the conflicting
 223 requirements of heterogeneous application traffic classes. Fourth, no published
 224 protocol uses RL to learn which routing paradigm (proactive, reactive, or hybrid)
 225 to use for each packet based on a joint context vector of urgency, energy,
 226 and link stability. Fifth, no RL-based WSN routing protocol incorporates a
 227 safety constraint that guarantees latency bounds for critical traffic regardless
 228 of the learned policy — a requirement for safety-critical IoT deployments.

229 3. System Model

230 3.1. Network Topology

231 Consider a WSN comprising N heterogeneous sensor nodes deployed
 232 randomly over a square area of side length A m according to a uniform
 233 distribution. Let $\mathcal{N} = \{1, 2, \dots, N\}$ denote the set of all nodes. A base
 234 station (BS) is located at the centre of the deployment area, and a secondary
 235 sink node is positioned at the right boundary, consistent with the multi-
 236 sink topology in our prior work [27]. Both the BS and sink are assumed
 237 to have unconstrained energy. Each sensor node $i \in \mathcal{N}$ is characterised by
 238 a fixed position (p_i^x, p_i^y) and a heterogeneous initial energy E_0^i drawn from
 239 a uniform distribution $E_0^i \sim \mathcal{U}(E_{\min}, E_{\max})$, reflecting the reality of diverse
 240 sensor hardware in smart city deployments [40].

241 A node i is considered alive if its residual energy $E_i(t) > 0$ at simulation
 242 round t , and dead otherwise. Dead nodes neither transmit nor receive. The
 243 network is considered operational as long as at least 10% of nodes remain
 244 alive.

245 3.2. Traffic Model

246 Smart city IoT deployments generate heterogeneous application traffic
 247 with fundamentally different QoS requirements. This paper models three
 248 traffic classes simultaneously:

- 249 • **Class A — Emergency traffic:** Generated by safety-critical sensors
 250 such as fire detectors, intrusion alarms, and accident monitors. Modelled
 251 as a Poisson process with arrival rate λ_A . Strict latency requirement:
 252 end-to-end delay must not exceed 100 ms.

- 253 • **Class B — Periodic telemetry:** Generated by environmental sensors
 254 reporting temperature, humidity, air quality, and energy consumption.
 255 Modelled as a deterministic periodic process with one packet per minute.
 256 Tolerates delays up to 30 seconds.
- 257 • **Class C — Event-driven uploads:** Generated by motion detectors,
 258 occupancy sensors, and camera triggers. Modelled as a Poisson process
 259 with arrival rate λ_C . Best-effort delivery with no hard latency constraint.

260 Each packet carries a class tag $U \in \{A, B, C\}$ in its header, which is read
 261 by the CARL-WSN context classifier at the transmitting node before routing
 262 decisions are made.

263 3.3. Channel and Energy Model

264 Wireless communication is modelled using the first-order radio energy
 265 model [41], which captures the dominant sources of energy consumption in
 266 sensor nodes: electronics and amplification. The energy consumed by node i
 267 to transmit an L -bit packet to a receiver at distance d is:

$$E_{tx}(L, d) = \begin{cases} L \cdot E_{elec} + L \cdot \varepsilon_{fs} \cdot d^2 & d < d_0 \\ L \cdot E_{elec} + L \cdot \varepsilon_{mp} \cdot d^4 & d \geq d_0 \end{cases} \quad (1)$$

268 where $d_0 = \sqrt{\varepsilon_{fs}/\varepsilon_{mp}} = 87.71$ m is the crossover distance between free-
 269 space and multipath propagation models. The energy consumed to receive
 270 an L -bit packet is:

$$E_{rx}(L) = L \cdot E_{elec} \quad (2)$$

271 The residual energy of node i after a transmission round is:

$$E_i(t+1) = E_i(t) - E_{tx}(L, d) - E_{rx}(L) \quad (3)$$

272 subject to $E_i(t) \geq 0$.

273 3.4. Link Reliability and Retransmission Model

274 Unlike idealised evaluations that assume guaranteed delivery, all protocols
 275 in this study transmit over a distance-dependent lossy channel, so that packet
 276 delivery is an emergent outcome rather than an assumption. The probability
 277 that an L -bit frame is successfully received over a link of length d is modelled

278 by a logistic packet-reception-ratio (PRR) function in the transitional-region
 279 form of Zuniga and Krishnamachari [42]:

$$\text{PRR}(d) = \frac{1}{1 + \exp(\alpha_{ch}(d - \beta_{ch}))} \quad (4)$$

280 where $\beta_{ch} = 90$ m is the midpoint distance at which the reception probabil-
 281 ity falls to one half, and $\alpha_{ch} = 0.18$ controls the sharpness of the transitional
 282 region. Links shorter than β_{ch} are highly reliable, while reliability degrades
 283 steeply beyond it.

284 Each hop employs a stop-and-wait automatic-repeat-request (ARQ)
 285 scheme consistent with IEEE 802.15.4 [43]: a frame is retransmitted up to
 286 $R_{\max} = 3$ times, and the hop succeeds if any attempt is acknowledged. A
 287 packet is delivered to the base station only if *every* hop on its multi-hop path
 288 succeeds within its retransmission budget; otherwise it is counted as lost.
 289 Each transmission attempt — successful or not — consumes the energy in
 290 (1)–(2), so retransmissions over marginal links are charged their true energy
 291 cost. To ensure a fair, paired comparison, the per-link, per-attempt channel
 292 outcomes are generated from a common random-number stream keyed on the
 293 seed, round, and link, so that every protocol experiences an identical channel
 294 realisation for a given seed.

295 End-to-end latency is likewise emergent: the latency of a delivered packet
 296 is the sum over its hops of the per-hop transmission and processing time plus
 297 the timeout penalty of any failed attempts, rather than a fixed per-mode
 298 constant.

299 3.5. Link Stability Model

300 To capture dynamic link quality variations, CARL-WSN maintains a per-
 301 node link stability score $L_s^i(t) \in [0, 1]$, computed as an Exponential Weighted
 302 Moving Average (EWMA) of the ACK success history over a sliding window
 303 of W most recent transmissions:

$$L_s^i(t) = \sum_{k=0}^{W-1} w_k \cdot a_k^i \quad (5)$$

304 where $a_k^i \in \{0, 1\}$ is the ACK outcome of the k -th most recent transmission
 305 from node i , and $w_k = \alpha_{\text{ewma}}(1 - \alpha_{\text{ewma}})^k/Z$ are normalised exponentially
 306 decaying weights with smoothing factor $\alpha_{\text{ewma}} = 0.5$ and normalisation
 307 constant $Z = \sum_{k=0}^{W-1} (1 - \alpha_{\text{ewma}})^k$, computed over a window of $W = 30$

recent transmissions. These values follow the WMEWMA link estimator of Woo and Culler [44]. The score is updated online from the real per-hop acknowledgement outcomes produced by the ARQ mechanism of Section 3.

3.6. Energy-Aware Relay Selection Model

CARL-WSN introduces an energy-aware relay selection mechanism that extends network lifetime by preferring next-hop nodes with higher residual energy. When forwarding a packet, node i selects the next-hop relay j^* from its set of alive neighbours $\mathcal{N}_i = \{j : d_{ij} \leq R_{tx}, E_j(t) > 0\}$ by maximising a composite score:

$$j^* = \arg \max_{j \in \mathcal{N}_i} \left[\left(w_p \cdot \frac{d_{iBS} - d_{jBS}}{d_{iBS}} + w_e \cdot \frac{E_j(t)}{E_0^j} \right) \cdot \text{PRR}(d_{ij}) \right] \quad (6)$$

where d_{iBS} and d_{jBS} are the distances from nodes i and j to the base station respectively, $w_p = 0.4$ is the progress weight, and $w_e = 0.6$ is the energy weight. The first term encourages geographic progress toward the BS, while the second term favours relays with higher residual energy ratios, distributing the forwarding load away from energy-depleted nodes. The composite score is further weighted by the link reliability $\text{PRR}(d_{ij})$ from (4), so that the relay metric prefers next hops that are simultaneously closer to the BS, higher in energy, and reliably reachable. Only candidates with positive progress ($d_{jBS} < d_{iBS}$) are considered, ensuring packets always advance toward the BS.

3.7. Critical-Node Packet Deferral

When a node's residual energy ratio falls below a critical threshold $E_r^i < T_{defer}$, CARL-WSN defers transmission of non-urgent packets (Class B and Class C) to preserve the node's remaining energy. To prevent unbounded deferral — which would indefinitely withhold traffic from a near-depleted node that never fully dies — deferral is bounded: a node may defer at most $\delta_{\max}$ consecutive packets, after which it is forced to transmit. Deferred packets incur only a minimal listening energy cost of $0.1 \cdot L \cdot E_{elec}$ and are counted as offered-but-not-delivered, so deferral trades packet delivery for extended lifetime. The operating point used in this paper, $T_{defer} = 0.05$ and $\delta_{\max} = 3$, is selected from the sensitivity analysis in Section 6.10 to retain the lifetime gain at a modest delivery cost. Class A emergency packets are never deferred regardless of node energy state — the safety constraint takes absolute priority. This deferral mechanism is a unique capability of urgency-aware protocols:

baseline protocols that do not distinguish traffic classes cannot selectively defer low-priority traffic to extend node lifetime.

3.8. Performance Metrics

Eight metrics are used to evaluate protocol performance:

Network lifetime: The First Node Death round $r_{FND} = \min\{t : \exists i, E_i(t) = 0\}$ and the Half Node Death round $r_{HND} = \min\{t : |\{i : E_i(t) = 0\}| \geq N/2\}$.

Packet Delivery Ratio:

$$PDR = \frac{D_{delivered}}{D_{total}} \quad (7)$$

Energy balance (Gini coefficient):

$$G = \frac{2 \sum_{i=1}^N i \cdot c_i}{N \sum_{i=1}^N c_i} - \frac{N+1}{N} \quad (8)$$

where c_i is the energy consumed by node i , sorted in ascending order ($c_1 \leq c_2 \leq \dots \leq c_N$). A lower G indicates more uniform energy consumption.

Routing overhead:

$$\Omega = \frac{C_{ctrl}}{D_{total}} \quad (9)$$

where C_{ctrl} is the total number of routing control packets generated during the simulation.

Additional metrics: Mean Class A latency, average energy consumed per round, and network throughput (packets delivered per round).

4. CARL-WSN Protocol Design

4.1. Design Motivation

Existing RL-based WSN routing protocols apply Q-Learning within a single routing paradigm. RLCR [8] uses Q-Learning for cluster head selection and inter-cluster forwarding — both decisions operate within the clustering paradigm. FQ-UCR [9] similarly applies Q-Learning only to select the next forwarding cluster head. In both protocols, the fundamental routing mechanism (clustering with multi-hop inter-cluster relay) never changes regardless of traffic urgency.

366 CARL-WSN addresses a categorically different problem: *which routing*
 367 *paradigm should be used for each packet?* The RL agent learns to select
 368 between proactive, reactive, and hybrid routing — three fundamentally differ-
 369 ent mechanisms with distinct latency, overhead, and energy characteristics —
 370 based on a per-packet context vector that includes application urgency class.
 371 This is routing paradigm switching, not routing optimisation within a fixed
 372 paradigm.

373 4.2. The Context Vector

374 For each packet to be transmitted by node i of class U , the context vector
 375 $\mathbf{C}_i = (U, E_r^i, L_s^i)$ is computed, where:

- 376 • $U \in \{A, B, C\}$ is the urgency class read directly from the packet header.
 377 Zero additional computation is required.
- 378 • $E_r^i = E_i(t)/E_0^i \in [0, 1]$ is the normalised residual energy ratio. $E_r^i = 1$
 379 indicates full energy; $E_r^i \rightarrow 0$ indicates near-depletion.
- 380 • $L_s^i \in [0, 1]$ is the link stability score computed using (5).

381 4.3. MDP Formulation

382 The routing paradigm selection problem is formulated as a Markov Decision
 383 Process $(\mathcal{S}, \mathcal{A}, R, \gamma)$ where each packet transmission constitutes one decision
 384 step.

385 4.3.1. State Space

386 The continuous context vector is discretised into a finite state space
 387 suitable for tabular Q-Learning:

- 388 • Urgency class U : 3 values — A (emergency), B (telemetry), C (event-
 389 driven).
- 390 • Residual energy level: 3 levels — low ($E_r^i < 0.3$), mid ($0.3 \leq E_r^i < 0.7$),
 391 high ($E_r^i \geq 0.7$).
- 392 • Link stability level: 2 levels — unstable ($L_s^i < 0.7$), stable ($L_s^i \geq 0.7$).

Table 2: CARL-WSN State Space: All 18 States

State	Real-world meaning	U	E_r level	L_s level
1–2	Emergency, low energy, unstable/stable link	A	low	unstable/stable
3–4	Emergency, moderate energy, unstable/stable	A	mid	unstable/stable
5–6	Emergency, healthy node, unstable/stable	A	high	unstable/stable
7–8	Telemetry, low energy, unstable/stable	B	low	unstable/stable
9–10	Telemetry, moderate energy, unstable/stable	B	mid	unstable/stable
11–12	Telemetry, healthy node, unstable/stable	B	high	unstable/stable
13–14	Best-effort, low energy, unstable/stable	C	low	unstable/stable
15–16	Best-effort, moderate energy, unstable/stable	C	mid	unstable/stable
17–18	Best-effort, healthy node, unstable/stable	C	high	unstable/stable

393 The total state space is $|\mathcal{S}| = 3 \times 3 \times 2 = 18$ states. Each state is encoded
 394 as a linear index:

$$s = (U_{num} - 1) \times 6 + (E_{level} - 1) \times 2 + L_{level} \quad (10)$$

395 where $U_{num} \in \{1, 2, 3\}$, $E_{level} \in \{1, 2, 3\}$, and $L_{level} \in \{1, 2\}$. Table 2
 396 enumerates all 18 states with their physical interpretations.

397 4.3.2. Action Space

398 The agent selects one of three routing paradigms per packet:

$$\mathcal{A} = \{\text{proactive, reactive, hybrid}\} \quad (11)$$

399 The Q-table has dimensions $18 \times 3 = 54$ entries, requiring only $54 \times 4 = 216$
 400 bytes of storage in single-precision floating point — deployable on any sensor
 401 platform with at least 1 KB of RAM.

402 4.3.3. Reward Function

403 After selecting action a in state s , the agent observes the routing outcome
 404 and receives a class-dependent multi-objective reward:

$$R(s, a) = w_1^U \cdot R_{lat} + w_2^U \cdot R_{oh} + w_3^U \cdot R_{en} \quad (12)$$

405 where the three reward components are:

$$R_{lat} = \begin{cases} -1 & \text{if } U = A \text{ and } \ell > 100 \text{ ms} \\ 1 - \ell / \ell_{\max}^U & \text{otherwise} \end{cases} \quad (13)$$

$$R_{oh} = 1 - \min\left(\frac{c_{ctrl}}{n_{nb} \cdot 0.20}, 1\right) \quad (14)$$

$$R_{en} = \min\left(\frac{E_r^{after}}{E_r^{before}}, 1\right) \quad (15)$$

Here ℓ is the observed end-to-end latency, $\ell_{\max}^U$ is the class-specific latency tolerance (100 ms for Class A, 30 s for Class B, ∞ for Class C), c_{ctrl} is the number of control packets generated, n_{nb} is the neighbour count, and E_r^{before} and E_r^{after} are the residual energy ratios before and after transmission.

The class-dependent weight vectors are:

$$(w_1^U, w_2^U, w_3^U) = \begin{cases} (0.7, 0.15, 0.15) & U = A \text{ (latency dominates)} \\ (0.2, 0.3, 0.5) & U = B \text{ (energy dominates)} \\ (0.1, 0.3, 0.6) & U = C \text{ (energy dominates)} \end{cases} \quad (16)$$

This formulation ensures that the RL agent learns different routing preferences for different traffic classes: Class A packets are penalised heavily for latency violations, while Class B and Class C packets are rewarded primarily for energy preservation. The weight vectors encode the fundamental QoS distinction between emergency and non-emergency IoT traffic.

4.3.4. Q-Learning Update Rule

The Q-table is updated after each packet transmission using the standard Q-Learning rule:

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[R(s, a) + \gamma \max_{a'} Q(s', a') - Q(s, a) \right] \quad (17)$$

where $\alpha = 0.1$ is the learning rate, $\gamma = 0.9$ is the discount factor, s' is the next state observed after packet delivery, and $\max_{a'} Q(s', a')$ is the maximum expected future reward from the next state.

Action selection follows an ε -greedy policy:

$$a = \begin{cases} \text{random action from } \mathcal{A} & \text{with probability } \varepsilon \\ \arg \max_a Q(s, a) & \text{with probability } 1 - \varepsilon \end{cases} \quad (18)$$

The exploration rate decays exponentially:

$$\varepsilon_{t+1} = \max(\varepsilon_t \cdot \delta, \varepsilon_{\min}) \quad (19)$$

424 with initial $\varepsilon_0 = 0.1$, decay factor $\delta = 0.998$, and minimum $\varepsilon_{\min} = 0.01$.
 425 This schedule ensures sufficient exploration in early rounds while converging
 426 to a near-greedy exploitation policy within approximately 1150 rounds.

427 4.4. Safety Constraint

428 A critical design requirement for safety-critical IoT deployments is that
 429 emergency traffic must *never* be delayed by route discovery, regardless of
 430 what the RL agent has learned. CARL-WSN enforces this through a hard
 431 constraint that overrides the agent’s policy:

$$\text{If } U = A : \quad a = \text{proactive} \quad (\text{unconditional}) \quad (20)$$

432 This constraint is applied *before* the ε -greedy selection in (18), ensuring
 433 that Class A packets are always routed via proactive mode with pre-built
 434 routing tables, guaranteeing sub-100 ms delivery at all times. The Q-table
 435 entries for states 1–6 (all Class A states) are never updated by the agent —
 436 proactive routing for emergencies is a design invariant, not a learned behaviour.

437 This separation between learned policy (Class B and C) and hardcoded
 438 safety guarantee (Class A) is a distinguishing feature of CARL-WSN. Neither
 439 RLCR [8] nor FQ-UCR [9] implements traffic-class-specific safety constraints,
 440 because neither protocol differentiates traffic by urgency class.

441 4.5. Routing Modes

442 The three routing paradigms available to the RL agent are:

443 4.5.1. Proactive Mode

444 Inspired by DSDV [3], proactive mode maintains routing tables through
 445 periodic broadcasts every round. When a packet requires transmission, the
 446 next hop toward the BS is selected using the energy-aware relay selection
 447 mechanism defined in (6), combining geographic progress with residual energy
 448 preference. The energy overhead includes both data transmission cost and
 449 a table maintenance overhead of 10% of the data packet energy per round.
 450 This mode is mandatory for Class A emergency traffic and is also selected by
 451 the RL agent when link instability makes reactive route discovery unreliable.

452 4.5.2. *Reactive Mode*

453 Inspired by AODV [4], reactive mode initiates route discovery only when
 454 a packet needs to be sent. A Route Request (RREQ) is flooded to neighbours
 455 within transmission range, and a Route Reply (RREP) is returned along the
 456 reverse path with energy-aware relay selection applied at each hop. This
 457 eliminates periodic broadcast overhead at the cost of an on-demand route-
 458 discovery delay on a cache miss. Discovered routes are cached and reused
 459 while valid, so the discovery cost is amortised over subsequent packets to the
 460 same destination; the per-packet latency is therefore emergent — the sum of
 461 per-hop transmission times plus the discovery round-trip on the fraction of
 462 packets that miss the cache — rather than a fixed per-mode constant. The RL
 463 agent learns to select reactive mode primarily for Class B and Class C traffic
 464 on healthy nodes with stable links, where the energy saving from avoiding
 465 table maintenance outweighs the latency cost.

466 4.5.3. *Hybrid Mode*

467 Extending ZRP [7], hybrid mode maintains proactive routes within a
 468 dynamically-sized zone and uses reactive inter-zone routing beyond the zone
 469 boundary. The proactive zone radius adapts based on the node’s residual
 470 energy ratio:

$$r_{zone}^i = \begin{cases} 120 \text{ m} & E_r^i \geq 0.6 \text{ (healthy)} \\ 80 \text{ m} & 0.3 \leq E_r^i < 0.6 \text{ (moderate)} \\ 40 \text{ m} & E_r^i < 0.3 \text{ (critical)} \end{cases} \quad (21)$$

471 The RL agent learns to select hybrid mode primarily for energy-critical
 472 nodes carrying non-emergency traffic, where the reduced zone radius limits
 473 broadcast overhead while maintaining route availability within the immediate
 474 neighbourhood.

475 4.6. *Complete Per-Packet Routing Procedure*

476 Algorithm 1 presents the complete per-packet routing procedure, inte-
 477 grating the context vector computation, safety constraint, ε -greedy action
 478 selection, energy-aware routing, packet deferral, and Q-table update.

479 4.7. *Overhead Analysis*

480 The per-packet control overhead ω of CARL-WSN adapts to the selected
 481 mode:

$$\omega = \begin{cases} n_{nb} \cdot 0.08 & \text{proactive (Class A)} \\ n_{nb} \cdot 0.05 & \text{reactive (Class B/C)} \\ n_{nb} \cdot 0.06 & \text{hybrid} \end{cases} \quad (22)$$

where n_{nb} is the number of one-hop neighbours within 100 m range. By selecting reactive mode for the majority of Class B and C traffic when conditions permit, CARL-WSN avoids the continuous broadcast overhead that makes purely proactive protocols (DSDV: $\omega = n_{nb} \cdot 0.20$ per packet) energy-intensive.

4.8. Computational Complexity

The context classifier operates in $O(1)$ time per packet: urgency class U is read from the packet header in constant time, E_r^i is a single division operation, L_s^i is a fixed-length dot product over $W = 30$ values, state indexing requires three integer operations, and the Q-table lookup is a single array access. The Q-Learning update in (17) requires one multiplication, one addition, and one max operation over 3 values — all $O(1)$. The total per-packet computational overhead is under 0.1 ms on any modern sensor processor. The memory footprint comprises the Q-table (216 bytes), the ACK history buffer (80 bytes), and the state index computation variables (12 bytes) — a total of 308 bytes, deployable on TelosB (10 KB RAM), MicaZ (4 KB RAM), and Arduino-class devices without modification.

5. Simulation Setup

5.1. Network Model

The simulation models a smart city IoT-WSN deployment comprising $N = 100$ heterogeneous sensor nodes randomly distributed over a $200 \times 200 \text{ m}^2$ area using a uniform random distribution. A base station is positioned at the centre of the deployment area at coordinates (100, 100) m, and a secondary sink is placed at (200, 100) m, as illustrated in Fig. 1, consistent with the multi-sink topology established in our prior work [27]. Nodes are initialised with heterogeneous energy levels drawn from a uniform distribution $E_0 \sim \mathcal{U}(0.5, 2.0)$ J. MATLAB-based simulation is a well-established and widely accepted methodology for WSN routing protocol evaluation, as demonstrated by recent published studies [32, 8, 45].

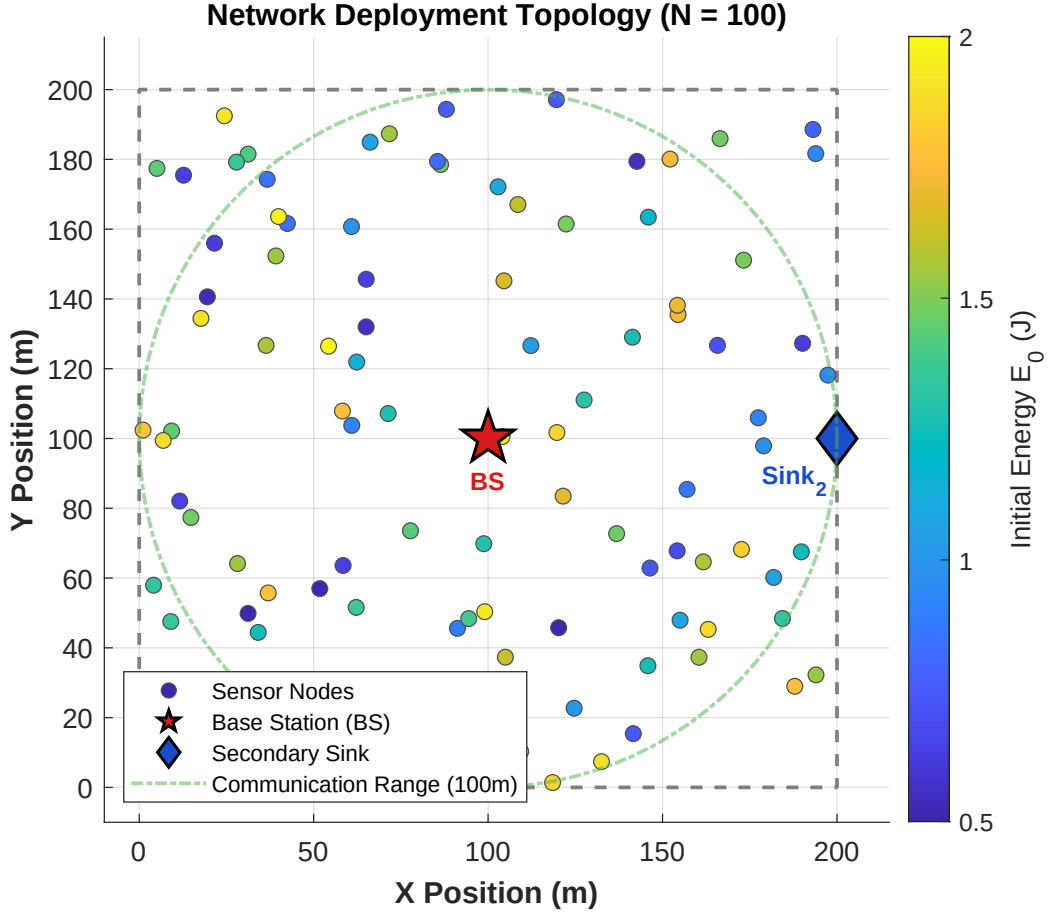


Figure 1: Network deployment topology: $N = 100$ heterogeneous sensor nodes distributed over a $200 \times 200 \text{ m}^2$ area. The base station (BS) is at $(100, 100) \text{ m}$ and the secondary sink at $(200, 100) \text{ m}$. Node colour indicates initial energy $E_0 \sim \mathcal{U}(0.5, 2.0) \text{ J}$. The dashed circle shows the 100 m communication range around the BS.

511 5.2. Energy Model

512 The first-order radio energy model [41] is adopted, following (1) and (2)
 513 as defined in Section 3, with parameters listed in Table 3.

514 5.3. Traffic Scenarios

515 Three IoT application traffic classes are simulated simultaneously:

- 516 • **Class A (Emergency):** Poisson arrivals, $\lambda_A = 0.5 \text{ packets/s}$, latency
 517 requirement $< 100 \text{ ms}$.

- 518 • **Class B (Periodic Telemetry):** One packet per minute, energy-
519 efficient delivery.
- 520 • **Class C (Event-Driven):** Poisson arrivals, $\lambda_C = 0.1$ packets/s, best-
521 effort delivery.

522 Four traffic scenarios are evaluated: Class A only, Class B only, Class C
523 only, and Mixed. In the mixed-traffic scenario, packets are generated with
524 proportions of 20%, 50%, and 30% for Classes A, B, and C respectively. In all
525 four scenarios, the total traffic volume per round is held constant — only the
526 class composition varies. This experimental design ensures that performance
527 differences across scenarios are attributable to context-aware routing decisions,
528 not to differences in offered load. Consequently, baseline protocols that do not
529 differentiate traffic classes produce identical performance across all scenarios,
530 while CARL-WSN’s metrics vary as the RL agent selects different routing
531 modes for different class compositions. These proportions reflect a realistic
532 smart city IoT deployment in which emergency events are infrequent, periodic
533 environmental monitoring dominates network traffic, and event-driven sensing
534 forms an intermediate tier [26, 2].

535 5.4. *Simulation Parameters*

536 Table 3 summarises all simulation parameters used in this study, including
537 the Q-Learning hyperparameters for CARL-WSN.

538 5.5. *Experimental Protocol*

539 Each of the eight protocols is evaluated under all four traffic scenarios with
540 20 independent random seeds (42–61), giving $8 \times 4 \times 20 = 640$ independent
541 simulation runs. Crucially, all eight protocols — including the seven baselines —
542 are re-implemented on the common lossy-channel and ARQ model of Section 3,
543 so that packet delivery, latency, and energy consumption emerge from identical
544 physical assumptions rather than from protocol-specific idealisations or values
545 quoted from their original publications. For a given seed, the per-link channel
546 realisation is shared across all protocols, yielding a paired comparison on an
547 identical topology and channel.

548 To obtain unbiased lifetime statistics, each run proceeds until the network
549 is non-operational (fewer than 10% of nodes alive), subject to the safety cap
550 in Table 3; across all 640 runs no simulation reached this cap. Time-averaged
551 metrics (throughput and average energy per round) are computed over a

552 common horizon — the longest network-death round across all protocols for
 553 a given scenario and seed — so that protocols failing at different times are
 554 compared over an identical observation window.

555 5.6. Baseline Protocols

556 CARL-WSN is evaluated against seven baseline protocols spanning three
 557 categories:

558 **Classical routing paradigms:** (1) AODV [4] — reactive routing,
 559 (2) DSDV [3] — proactive routing, (3) ZRP [7] — hybrid routing. These
 560 represent the three paradigms between which CARL-WSN switches, enabling
 561 direct evaluation of whether context-aware RL-based mode selection outper-
 562 forms each individual paradigm operating in isolation. Because all three
 563 are re-implemented on the common data-plane of Section 5 — the same
 564 energy-aware forwarding substrate, lossy channel, and ARQ mechanism —
 565 they differ only in their control-plane behaviour: route-acquisition latency
 566 and control overhead. Consequently their lifetime, delivery, energy, and
 567 throughput results coincide by construction, and they are distinguished in
 568 our evaluation only by latency and overhead.

569 **Domain-specific baselines:** (4) EH-Routing [40] — energy-harvesting
 570 aware routing with highest-residual-energy relay selection, sharing the same
 571 heterogeneous energy model; (5) MSLBA [27] — multi-sink load-balanced
 572 routing, sharing the same dual-sink topology.

573 **Recent RL-based protocols:** (6) RLCR [8] — Q-Learning for cluster
 574 head selection and inter-cluster multi-hop routing, published in Ad Hoc
 575 Networks (2026); (7) FQ-UCR [9] — fuzzy logic for unequal cluster head
 576 selection combined with Q-Learning for inter-cluster forwarding, published in
 577 Entropy (2025). RLCR and FQ-UCR are included as direct RL competitors,
 578 enabling evaluation of whether CARL-WSN’s paradigm-switching approach
 579 outperforms RL applied within a fixed clustering paradigm.

580 The seven baselines were selected to provide comprehensive coverage
 581 across classical paradigms, domain-specific approaches, and recent RL-based
 582 methods. The context-aware works surveyed in Table 1 operate at different
 583 protocol layers (RPL parent selection, fog forwarding, opportunistic trust
 584 routing) and are therefore not directly comparable under the same WSN
 585 simulation framework.

586 5.7. Performance Metrics

587 Eight metrics are evaluated: (1) First Node Death round (FND), (2) Half
588 Node Death round (HND), (3) Packet Delivery Ratio (PDR), (4) Mean Class A
589 latency, (5) Average energy consumption per round, (6) Gini coefficient for
590 energy balance, (7) Routing overhead, and (8) Network throughput. All
591 results are reported as the mean across 20 independent runs (seeds 42–61).

592 6. Results and Analysis

593 Throughout this section, all eight protocols are evaluated under the unified
594 methodology of Section 5: each is executed on the same distance-dependent
595 lossy channel with per-hop acknowledgement and bounded retransmission,
596 over identical node deployments and channel realisations, so that observed
597 differences reflect routing behaviour rather than protocol-specific physical
598 assumptions. Unless stated otherwise, every reported value is the mean over
599 20 independent seeds (42–61), and error bars in all figures denote one standard
600 deviation across those seeds.

601 Statistical comparisons follow a two-stage procedure with a significance
602 threshold of $\alpha = 0.05$. For each metric, a one-way ANOVA first tests whether
603 the eight protocols differ overall; where this omnibus test is significant, Tukey’s
604 Honestly Significant Difference (HSD) post-hoc test identifies which specific
605 pairs of protocols differ. Consequently, a statement that CARL-WSN is
606 *significantly better* or *significantly worse* than a baseline on a given metric is
607 supported by a significant Tukey pairwise comparison, whereas describing two
608 protocols as *comparable* means the Tukey test found no significant difference
609 between them, even when their mean values differ. Descriptive rankings, such
610 as the highest mean value of a metric, are reported separately and do not by
611 themselves imply statistical significance.

612 6.1. Class A Emergency Packet Latency

613 Fig. 2 presents the mean Class A emergency packet latency under the
614 mixed-traffic scenario. CARL-WSN achieves a mean latency of 24.2 ms, a
615 **25.6% reduction** compared to AODV (32.5 ms). This validates the safety-
616 constraint design: Class A packets are always routed proactively, eliminating
617 the on-demand route-discovery delay that dominates AODV’s emergency
618 latency. All protocols remain well within the 100 ms emergency deadline.
619 One-way ANOVA confirms that latency differences across the eight protocols
620 are statistically significant ($p < 0.001$), and Tukey HSD post-hoc analysis

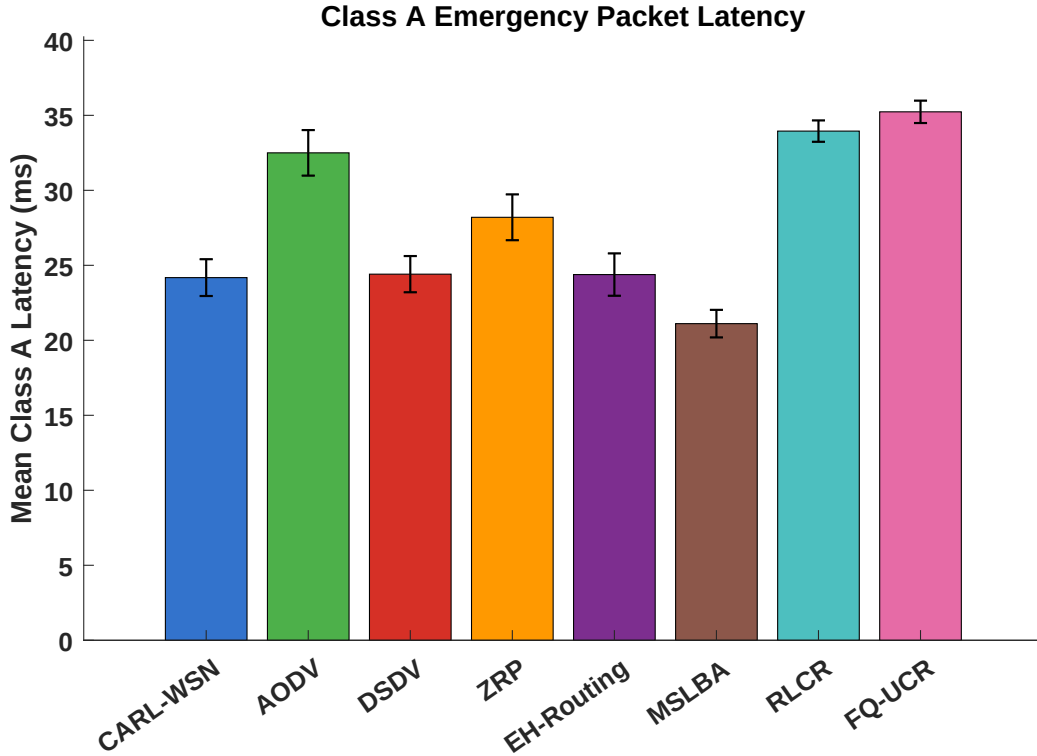


Figure 2: Mean Class A emergency packet latency under mixed traffic. CARL-WSN achieves a 25.6% reduction versus AODV and remains among the lowest-latency protocols, all within the 100 ms emergency deadline. Error bars show standard deviation across 20 independent seeds.

621 confirms CARL-WSN achieves significantly lower Class A latency than AODV,
 622 ZRP, RLCR, and FQ-UCR (all $p < 0.001$).

623 CARL-WSN does not achieve the single lowest latency: MSLBA attains
 624 21.1 ms through its dedicated dual-sink forwarding, and DSDV and EH-
 625 Routing are statistically comparable to CARL-WSN (≈ 24 ms), since all three
 626 avoid route-discovery delay. CARL-WSN’s contribution is therefore to deliver
 627 latency among the lowest of all protocols — and significantly below every
 628 reactive and learning-based baseline — while simultaneously adapting its
 629 routing overhead to traffic class (Section 6, Table 5), a property none of the
 630 lower-latency baselines possess.

6.2. Routing Overhead

CARL-WSN incurs a routing overhead of 0.53 control packets per data packet under mixed traffic (Fig. 3), substantially lower than all the flooding- and table-based protocols: a 96.5% reduction versus AODV (14.82), a 91.1% reduction versus ZRP (5.91), and a 77.2% reduction versus DSDV (2.31). It is also below the two domain-specific baselines EH-Routing and MSLBA (1.00 each). One-way ANOVA confirms the overhead differences are statistically significant ($p < 0.001$).

The overhead difference between CARL-WSN and AODV is the most pronounced. AODV’s reactive route discovery floods the network with RREQ packets on every cache miss, which under the high traffic load of one packet per node per round dominates its control budget. We note that control packets are accounted in the overhead metric of Equation (9) but are not charged against the radio energy model: the energy and lifetime results reflect data-plane transmissions only. This isolates control overhead as a distinct cost dimension and explains why protocols with very different overhead (e.g. AODV at 14.82 versus DSDV at 2.31) exhibit identical energy and lifetime under our model. CARL-WSN instead uses energy-aware geographic relaying with cached, on-demand discovery, avoiding network-wide floods entirely.

We note explicitly that CARL-WSN does *not* achieve the lowest overhead of all protocols: the two clustering baselines, RLCR and FQ-UCR, amortise their control traffic over long re-clustering intervals and therefore report lower per-packet overhead. Under a re-clustering-only accounting they incur approximately 0.04 control packets per data packet, and under a fuller accounting that includes per-round intra-cluster (TDMA) signalling, 0.10–0.18. Both values fall below CARL-WSN’s 0.53. This is an inherent advantage of the clustering paradigm’s steady-state structure, which CARL-WSN trades away in exchange for per-packet context adaptation and freedom from cluster-head energy hotspots (Section 6). CARL-WSN’s overhead advantage is thus specifically over the flooding- and table-based protocols, not over clustering.

6.3. Network Lifetime

Network lifetime is CARL-WSN’s strongest result. Table 4 presents the complete set of metrics under mixed traffic. CARL-WSN achieves the **highest first-node-death (FND) round of all eight protocols** at $FND = 418$, ahead of the next-best baselines RLCR (390), EH-Routing (380), MSLBA (362), and the classical flat-routing protocols AODV/DSDV/ZRP (327), and far ahead of FQ-UCR (123), whose unequal-clustering scheme depletes cluster

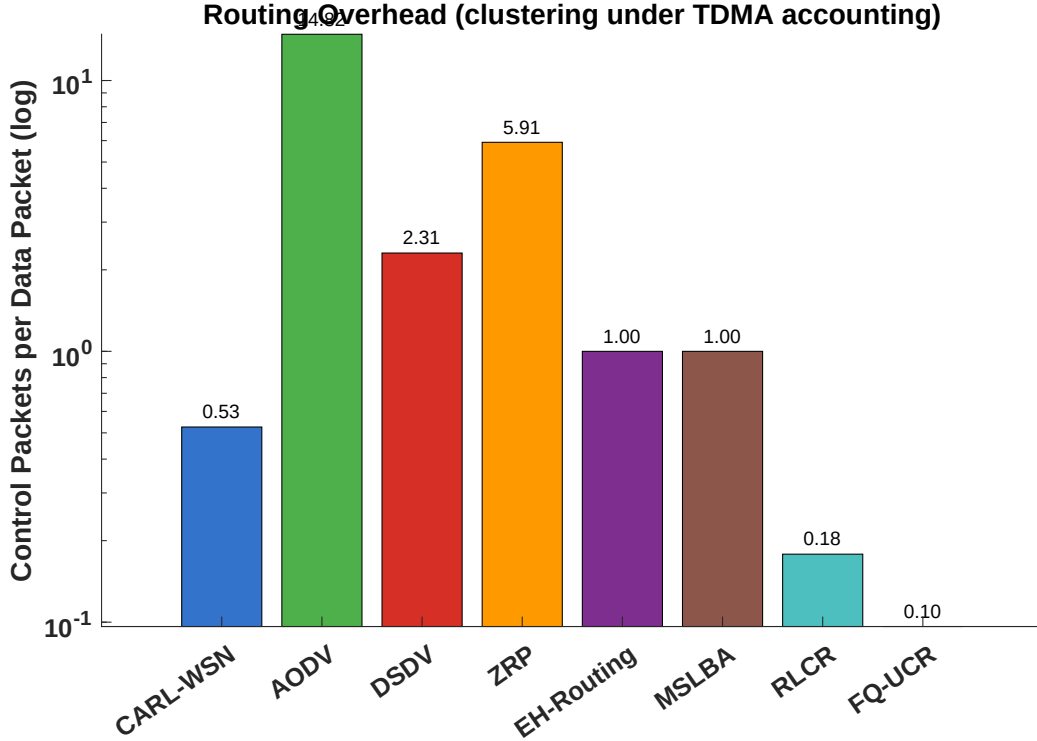


Figure 3: Routing overhead (control packets per data packet, log scale). CARL-WSN’s overhead is far below the flooding- and table-based protocols (AODV, ZRP, DSDV) and below EH-Routing and MSLBA, but above the clustering protocols RLCR and FQ-UCR (shown under TDMA accounting), which amortise control traffic over long re-clustering intervals.

heads near the base station early. This confirms that the combination of energy-aware relay selection in (6) and bounded critical-node packet deferral effectively delays the first energy depletion in the network.

On half-node-death (HND), CARL-WSN attains $HND = 2198$, the **second-highest** of all protocols, behind only MSLBA (2340) and ahead of EH-Routing (2126) and the classical protocols (1877). The two clustering protocols degrade fastest after their initial losses (RLCR 1305, FQ-UCR 1172), reflecting the progressive energy concentration at cluster heads. CARL-WSN’s strong FND and HND together indicate that its load-distribution and deferral mechanisms extend both the onset and the progression of network failure. One-way ANOVA confirms that FND and HND differ significantly across the eight protocols ($p < 0.001$). Tukey HSD analysis localises these differences: CARL-WSN’s

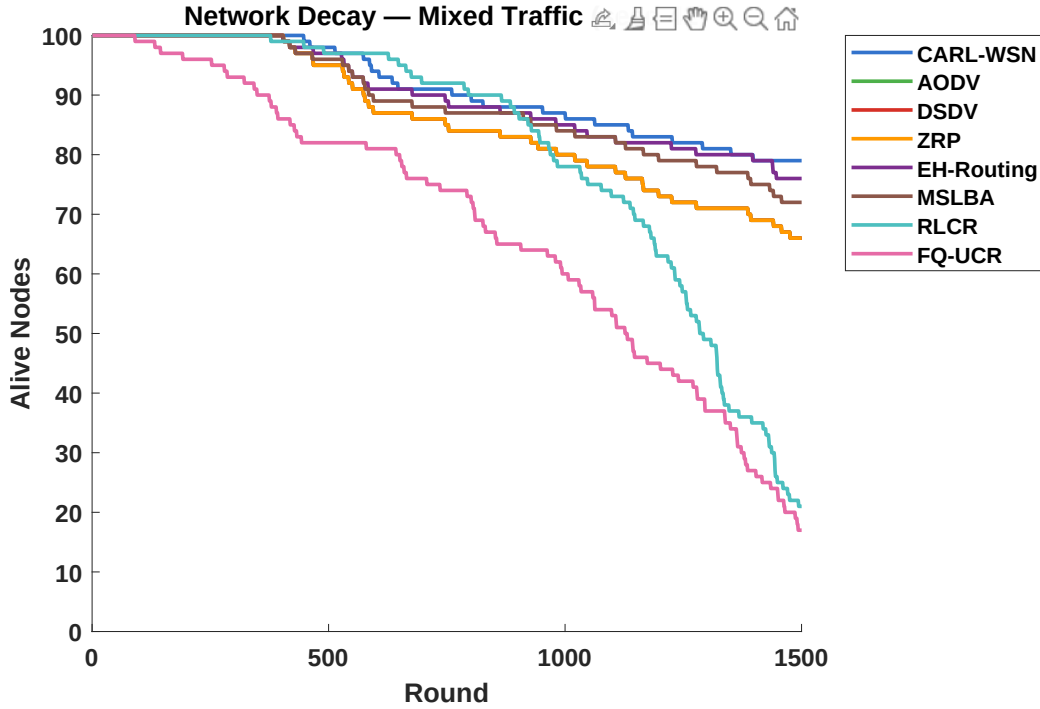


Figure 4: Number of alive nodes over time under mixed traffic (representative seed). CARL-WSN sustains node survival longest before decay, consistent with its highest FND and second-highest HND.

680 FND is significantly higher than the classical protocols (AODV/DSDV/ZRP)
681 and FQ-UCR, and its HND is significantly higher than the classical pro-
682 tocols, RLCR, and FQ-UCR. CARL-WSN attains the highest mean FND
683 and the second-highest mean HND overall, but its lifetime advantage over
684 EH-Routing, MSLBA, and RLCR is within seed-to-seed variation and is not
685 statistically significant; on lifetime, CARL-WSN is therefore best described
686 as leading the classical and clustering baselines while remaining comparable
687 to the energy-oriented baselines (EH-Routing, MSLBA).

688 Fig. 4 shows the alive-nodes-over-time curves under mixed traffic for a
689 representative seed. CARL-WSN sustains a high number of live nodes for the
690 longest period before decay, consistent with its leading FND and near-leading
691 HND.

692 6.4. Energy Efficiency and Balance

693 Average per-round energy consumption is statistically indistinguishable
 694 across all eight protocols (≈ 0.024 J per round; one-way ANOVA $p > 0.9$).
 695 This is expected: every protocol transmits over the same radio model with the
 696 same per-hop energy cost, so for a fixed offered load the total energy delivered
 697 per active round is essentially identical. Energy consumption is therefore not
 698 a differentiating metric in this study, and we report it for completeness rather
 699 than as a point of comparison.

700 Energy *balance*, measured by the Gini coefficient of per-node energy ex-
 701 penditure (Fig. 5), is tightly clustered across protocols: values range narrowly
 702 from 0.193 (MSLBA) to 0.203 (EH-Routing), with CARL-WSN at 0.200.
 703 The one-way ANOVA is marginally significant ($p = 0.048$), but this result is
 704 driven solely by the EH-Routing versus MSLBA pair; no pairwise comparison
 705 involving CARL-WSN is significant. We therefore make no energy-balance
 706 claim for CARL-WSN: its balance is statistically indistinguishable from every
 707 other protocol, indicating that its energy-aware relay selection maintains
 708 balance comparable to the baselines without concentrating consumption at
 709 particular nodes, and in particular without the cluster-head concentration
 710 that the clustering protocols must manage through rotation.

711 6.5. Throughput

712 Fig. 6 reports network throughput in packets delivered per round under
 713 mixed traffic, averaged over a common horizon so that protocols are compared
 714 over the same time window regardless of when their networks fail. CARL-
 715 WSN achieves 37.8 packets per round, comparable to the classical protocols
 716 (AODV/DSDV/ZRP at 36.7) and EH-Routing (38.2). MSLBA attains the
 717 highest throughput (48.5) by virtue of its dual-sink design, which shortens
 718 delivery paths. The two clustering protocols deliver the lowest throughput
 719 (RLCR 23.2, FQ-UCR 19.5); because throughput is averaged over a common
 720 horizon, their early network death penalises their delivered-per-round figure
 721 — a dead network delivers nothing for the remainder of the window.

722 CARL-WSN’s throughput is therefore mid-pack: it is not a through-
 723 put leader, but it sustains delivery comparable to the classical and energy-
 724 harvesting protocols while providing its lifetime and overhead advantages. Its
 725 slightly lower PDR (Section 6) is partly offset in this metric by its longer
 726 survival, which keeps more nodes generating and delivering traffic over the
 727 horizon.

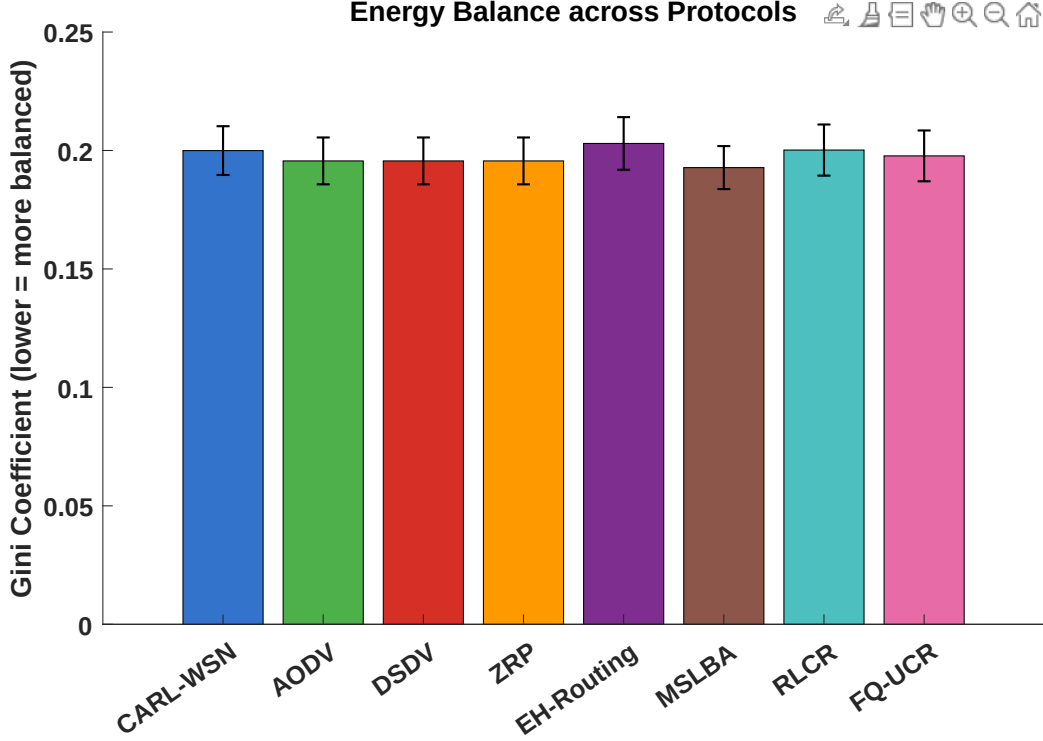


Figure 5: Gini coefficient of per-node energy consumption (lower is more balanced). Values are tightly clustered (0.193–0.203); the omnibus ANOVA is marginally significant ($p = 0.048$), driven solely by the EH-Routing versus MSLBA pair, and no comparison involving CARL-WSN (0.200) is significant.

6.6. Packet Delivery Ratio

Because every protocol is evaluated on a common distance-dependent lossy channel with per-hop acknowledgement and bounded retransmission, packet delivery is an emergent outcome rather than an idealised guarantee. Under mixed traffic the baselines achieve PDR in the range 91.5–96.9%, with the losses arising from transmissions over marginal links that exhaust their retransmission budget. CARL-WSN achieves 90.2% PDR — approximately five percentage points below the baseline group, and comparable to FQ-UCR (91.5%).

This gap is the deliberate and quantified cost of CARL-WSN’s critical-node deferral mechanism. When a node’s residual energy falls below the deferral threshold, it defers a bounded number of non-urgent (Class B/C) packets to preserve energy, and these deferred packets are counted as offered-but-

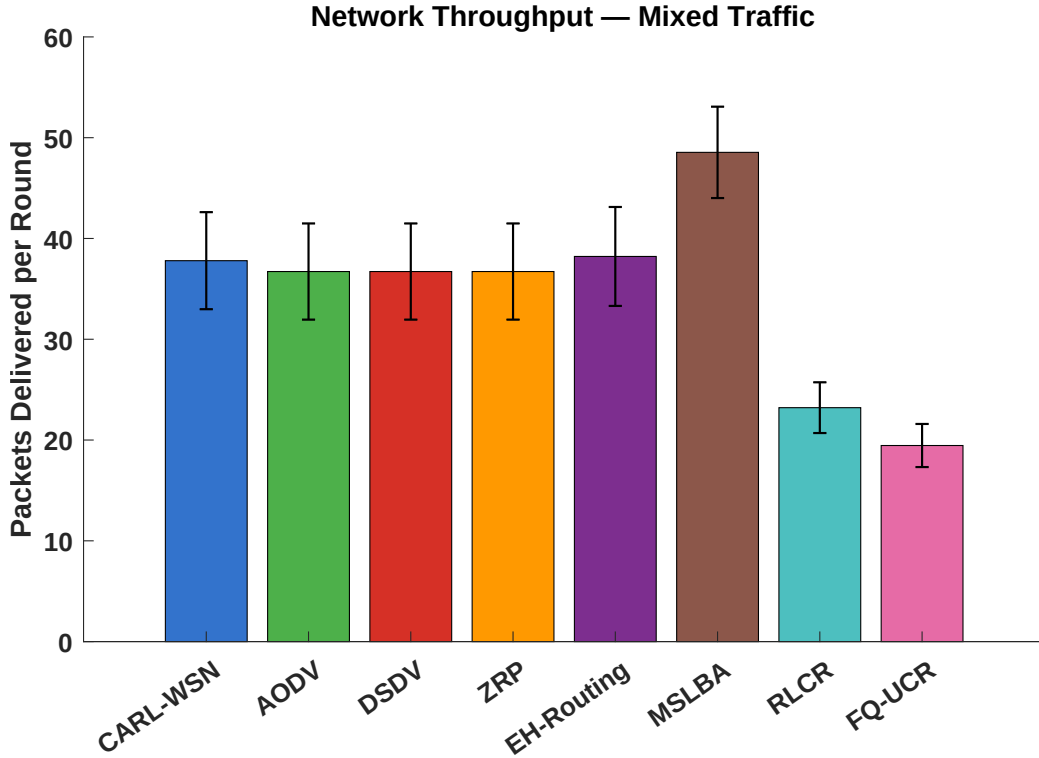


Figure 6: Network throughput (packets delivered per round) under mixed traffic, averaged over a common horizon. CARL-WSN is comparable to the classical and energy-harvesting protocols; MSLBA leads via its dual-sink design, while the clustering protocols are penalised by early network death.

not-delivered. Deferral is what produces CARL-WSN’s lifetime advantage; the PDR reduction is its direct counterpart. The operating point used here ($T_{\text{defer}} = 0.05$, at most three consecutive deferrals) was selected from the sensitivity analysis in Section 6.10 to keep this cost modest while retaining the lifetime gain. Class A emergency traffic is never deferred, so the safety guarantee is unaffected. The trade is thus explicit: CARL-WSN exchanges a small, bounded reduction in best-effort delivery for a significant extension of network lifetime.

6.7. Context-Awareness Validation

A defining property of CARL-WSN, absent from all seven baselines, is that its routing overhead adapts to the traffic-class composition of the workload. Table 5 reports CARL-WSN’s routing overhead in each single-class

753 scenario and under mixed traffic, alongside three representative baselines
 754 whose overhead is constant by construction.

755 CARL-WSN’s overhead varies more than three-fold across scenarios: 1.00
 756 control packet per data packet for Class A only, where the safety constraint
 757 forces every packet onto proactive routing with maintained neighbour state;
 758 0.32 for Class B and Class C only, where the agent predominantly selects
 759 reactive forwarding with cached, on-demand discovery; and 0.53 for the
 760 mixed workload, between the per-class extremes. The baselines, by contrast,
 761 generate identical overhead regardless of traffic composition, because they
 762 do not differentiate packets by urgency class. This empirically confirms
 763 that the RL agent learns class-specific routing behaviour — proactive for
 764 emergency traffic (higher overhead, lowest latency) and reactive for non-urgent
 765 traffic (lower overhead) — exactly the adaptation the class-dependent reward
 766 function in (12) was designed to produce.

767 6.8. *Q-Learning Convergence and Learned Policy*

768 Fig. 7 shows the maximum Q-table update magnitude ($\max |\Delta Q|$) per
 769 round on a logarithmic scale. The updates decrease from $\sim 10^0$ in the first
 770 rounds to $\sim 10^{-3}$ by round 1150, confirming that the Q-table converges to a
 771 stable policy well before half-node death (round 2198 under mixed traffic),
 772 while most nodes remain alive. After convergence, the agent exploits its
 773 learned policy with only occasional exploration ($\varepsilon_{\min} = 0.01$).

774 Fig. 8 visualises the learned routing policy as a heatmap over all 18 states,
 775 split by link stability, for a representative training seed. The policy exhibits
 776 the structure the reward design intends, rather than collapsing to a single
 777 mode:

- 778 • **Class A (emergency):** proactive is selected in every energy and link-
 779 stability state. This is imposed by the safety constraint, which forces
 780 proactive routing for Class A and excludes these states from Q-learning;
 781 the corresponding Q-values (Fig. 9, states 1–6) therefore remain at
 782 their initial zero, visually confirming that the emergency guarantee is
 783 structural rather than learned.
- 784 • **Class B (telemetry):** the agent learns a context-dependent mix.
 785 Moderate- and high-energy nodes on less stable links favour reactive
 786 forwarding, which avoids the cost of maintaining proactive tables, while
 787 moderate-energy nodes on stable links favour hybrid forwarding, a zone-
 788 limited compromise between proactive reach and reactive economy. This

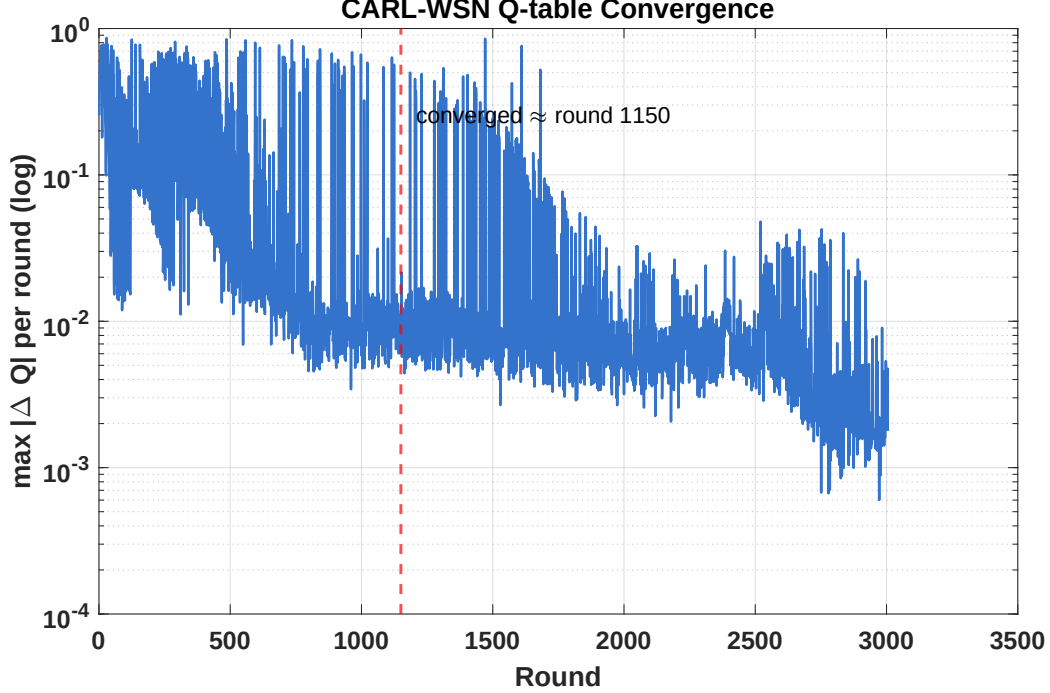


Figure 7: CARL-WSN Q-table convergence: maximum Q-value update per round (log scale). The agent converges to a stable policy within approximately 1150 rounds.

intermediate hybrid region — absent from any hand-designed rule — is discovered by the agent.

- **Class C (best-effort):** the agent converges to a consistent low-overhead choice across energy levels, reflecting the reward function’s heavy weighting of overhead and energy for the lowest-priority traffic.

The learned Q-values for Class B and Class C states (Fig. 9, states 7–18) are differentiated across the three actions, confirming that the agent explores the full action space and converges to context-specific preferences rather than defaulting to a single mode. We emphasise that the precise mode assignment in a given state may vary across training seeds, as it depends on exploration order; the robust and reproducible features are the unconditional proactive policy for Class A and the emergence of distinct, lower-overhead policies for non-urgent traffic.

Fig. 9 presents the learned Q-values for all 18 states. States 1–6 (Class A) show zero across all actions: under the safety constraint these states bypass

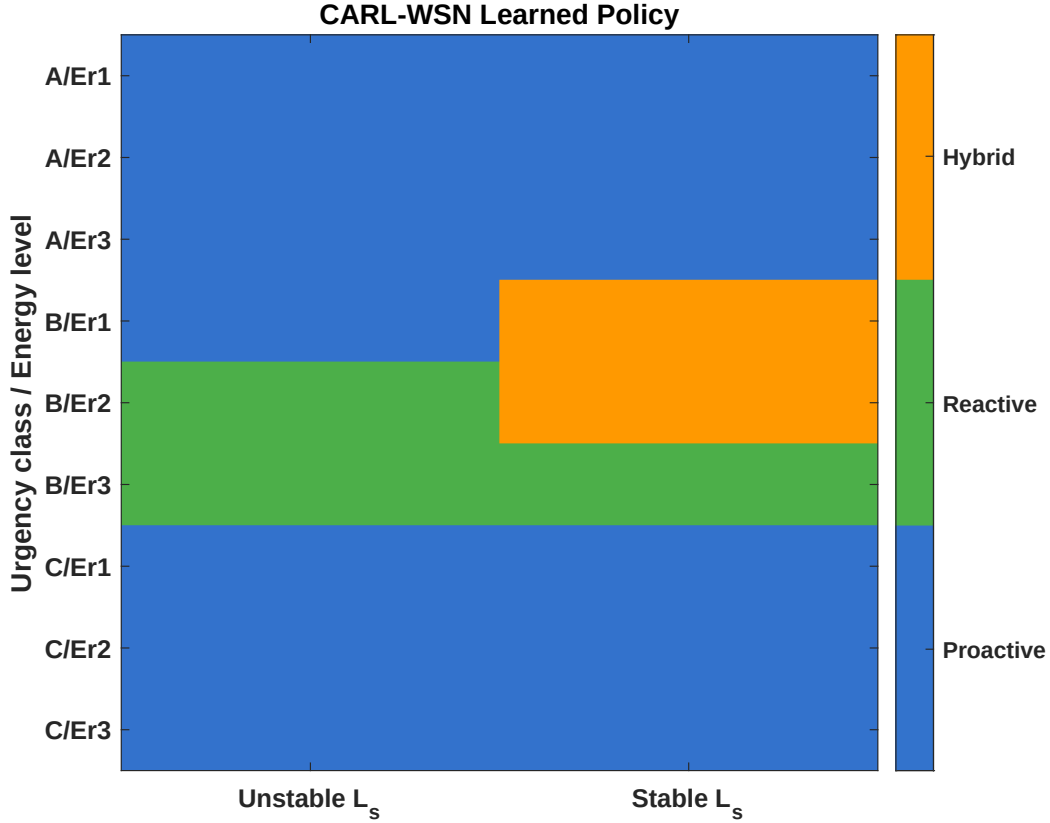


Figure 8: CARL-WSN learned routing policy across all 18 states. Left: unstable link ($L_s < 0.7$). Right: stable link ($L_s \geq 0.7$). Blue = proactive, green = reactive, orange = hybrid. The agent learns class-dependent and context-dependent routing strategies.

804 the Q-update entirely, so no values accrue and proactive routing is applied
 805 unconditionally. States 7–18 (Class B and C) show non-zero, differentiated
 806 values across the proactive, reactive, and hybrid actions, demonstrating that
 807 the agent has learned context-specific action preferences for the traffic classes
 808 it is permitted to optimise.

809 6.9. CARL-WSN vs RL Baselines

810 The three RL-based protocols occupy distinct regions of the design
 811 space. RLCR and FQ-UCR apply reinforcement learning *within* the cluster-
 812 ing paradigm — learning cluster-head selection and inter-cluster routing —
 813 whereas CARL-WSN applies it to per-packet paradigm selection on a flat

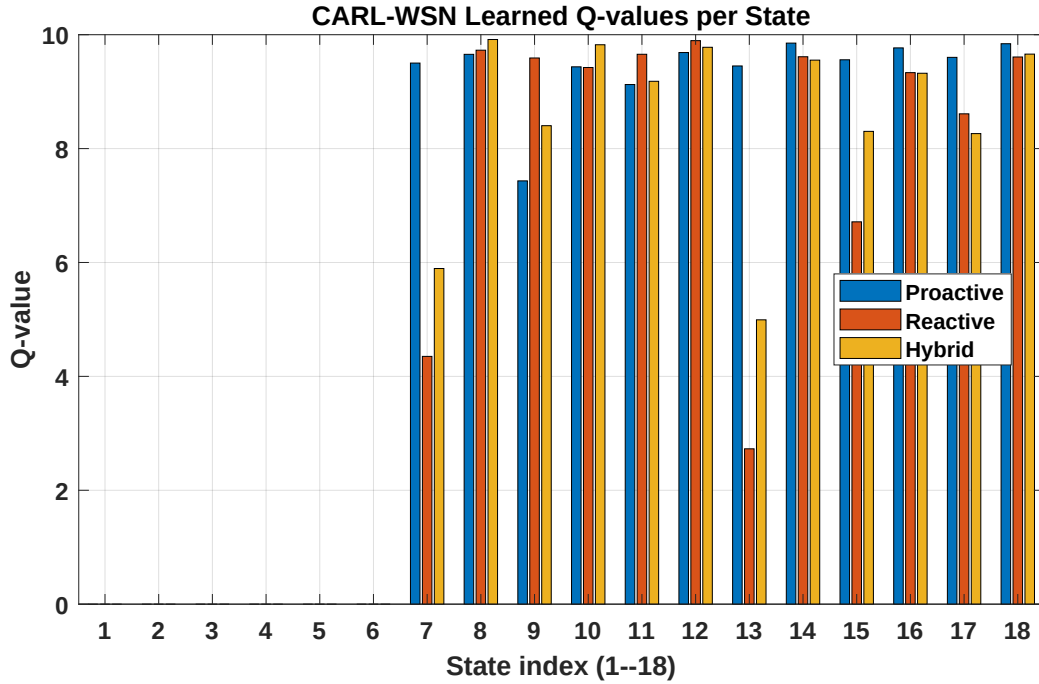


Figure 9: CARL-WSN learned Q-values for all 18 states. States 1–6 (Class A) are zero because the safety constraint excludes them from learning and forces proactive routing. States 7–18 (Class B/C) show differentiated values across the three routing modes, confirming context-dependent learning for non-emergency traffic.

814 topology. This produces a clear division of strengths rather than dominance
815 by any one protocol.

816 On network lifetime, CARL-WSN leads on the means: its FND (418)
817 exceeds RLCR’s (390) and greatly exceeds FQ-UCR’s (123), and its HND
818 (2198) is far above both clustering protocols (RLCR 1305, FQ-UCR 1172),
819 which degrade rapidly once cluster heads begin to fail. The HND gap over both
820 clustering protocols is statistically significant, as is the FND gap over FQ-UCR;
821 the FND difference from RLCR, however, is within seed variation and not
822 significant. On Class A latency, CARL-WSN (24.2 ms) is significantly lower
823 than both RLCR (34.0 ms) and FQ-UCR (35.2 ms), since neither clustering
824 protocol prioritises emergency traffic. On routing overhead, the ordering
825 reverses: the clustering protocols’ amortised control traffic (0.04–0.18 per
826 packet) is below CARL-WSN’s 0.53, an inherent benefit of their steady-state
827 cluster structure. On PDR, the three are close (CARL-WSN 90.2%, RLCR

828 94.1%, FQ-UCR 91.5%).

829 Table 6 summarises these pairwise differences. The picture is not one
830 of Pareto dominance in either direction: CARL-WSN trades higher routing
831 overhead for substantially better lifetime and emergency latency, while the
832 clustering protocols make the opposite trade. Which is preferable depends
833 on the deployment — CARL-WSN suits latency- and lifetime-critical hetero-
834 geneous traffic, whereas the clustering protocols suit overhead-constrained
835 homogeneous monitoring.

836 6.10. Sensitivity Analysis

837 To assess robustness, we vary CARL-WSN’s principal design parameters
838 one at a time around their default values, holding all else fixed, and re-run the
839 mixed-traffic scenario across the full set of 20 seeds. Two groups of parameters
840 are examined: the deferral policy (T_{defer} and $\max_{\text{defer}}$), which governs the
841 lifetime–delivery trade, and the model parameters that lack an exact value
842 in the literature (the relay weights, the channel shape parameters, and the
843 link-stability threshold).

844 6.10.1. Deferral Policy: the Lifetime–Delivery Frontier

845 The deferral parameters expose a direct trade between packet delivery
846 and network lifetime. Table 7 reports PDR and FND as T_{defer} and δ_{max} are
847 varied. Disabling deferral ($T_{\text{defer}} = 0$) yields the highest PDR (95.1%) at the
848 lowest FND (381), while aggressive deferral ($T_{\text{defer}} = 0.20$, $\delta_{\text{max}} = 10$) yields
849 the highest PDR (95.1%) at the lowest FND (381), while aggressive deferral
850 ($T_{\text{defer}} = 0.20$, $\delta_{\text{max}} = 10$) yields the highest FND (635) at the lowest PDR
851 (66.4%). The relationship is monotonic: deferring more non-urgent traffic at
852 near-depleted nodes extends lifetime at a proportional cost in delivery.

853 The operating point used throughout this paper, $T_{\text{defer}} = 0.05$ and
854 $\max_{\text{defer}} = 3$, is selected from this frontier as the point that retains a clear
855 lifetime advantage over all baselines (FND 418, the highest of any protocol)
856 while keeping the delivery cost modest (PDR 90.2%, within roughly five points
857 of the baseline group). More aggressive settings purchase additional lifetime
858 at a delivery cost we judge too steep for the marginal gain; for example,
859 raising T_{defer} to 0.15 extends FND by a further 107 rounds but reduces PDR
860 by almost 14 points. Deployments with different priorities can move along
861 this frontier by adjusting the two parameters.

6.10.2. Robustness to Model Parameters

Table 8 summarises the effect of the remaining parameters, reported as the percentage swing of each metric across the swept range. CARL-WSN is insensitive to the relay weight split (w_e/w_p): varying it from 0.4/0.6 to 0.8/0.2 changes PDR by under 1% and FND not at all, with only a mild effect on HND. The link-stability threshold T_L has no measurable effect, because the PRR-weighted relay metric keeps the large majority of links stable: across all forwarding decisions, the link-stability score falls below the 0.7 threshold in only 5.6% of cases, so varying the threshold within the 0.6–0.8 range alters few decisions. The channel sharpness parameter α likewise has negligible effect on PDR.

The one parameter with substantial effect is the PRR midpoint β , which sets the distance at which links begin to fail. Increasing β from 80 to 100 m improves PDR, FND, and HND monotonically, as expected: a more favourable channel reaches farther and delivers more. Crucially, because all eight protocols share the same channel model, β shifts every protocol together and does not alter CARL-WSN’s relative standing; it is a property of the deployment environment rather than of the protocol.

In summary, CARL-WSN’s behaviour is robust to its tunable design choices: delivery is essentially unchanged across the relay weights, the link-stability threshold, and the channel sharpness; lifetime shows only mild sensitivity to the channel sharpness ($\leq 15\%$ FND swing) and none to the other two; and all metrics vary with the channel midpoint only as the underlying physics dictates, equally for all protocols. The deferral policy is the one parameter with a deliberate, well-characterised effect, and the chosen operating point is justified by the frontier in Table 7.

6.11. Summary of Findings

Table 9 summarises CARL-WSN’s standing against the baseline group across the principal metrics under mixed traffic. The results support a focused set of claims. CARL-WSN delivers the **longest pre-failure lifetime** (highest FND, second-highest HND), the **lowest emergency latency among the reactive and learning-based protocols** (24.2 ms, 25.6% below AODV), and **routing overhead far below the flooding- and table-based protocols** (0.53 versus 2.31–14.82). These gains are obtained at a modest, bounded cost in packet delivery (90.2% PDR, ≈ 5 points below the baselines), while energy consumption is statistically comparable to every baseline. Energy balance shows no significant difference involving CARL-WSN, and throughput

899 is statistically comparable to the classical protocols and EH-Routing, above
900 the two clustering protocols, and below MSLBA.

901 These findings position CARL-WSN not as a universally dominant protocol
902 but as one that makes a specific, defensible trade: it extends network lifetime
903 and protects emergency-traffic latency, with low control overhead relative
904 to non-clustering protocols, in exchange for a small reduction in best-effort
905 delivery — a trade well suited to heterogeneous, latency- and lifetime-critical
906 IoT-WSN deployments.

907 7. Discussion

908 7.1. Lifetime Extension Without Clustering

909 The central finding of this study is that paradigm switching offers a
910 different operating point from paradigm optimisation, rather than dominating
911 it. CARL-WSN — which uses Q-Learning to switch *between* routing paradigms
912 on a flat topology — achieves the longest pre-failure lifetime (highest FND,
913 second-highest HND) and the lowest emergency latency among the learning-
914 based protocols, while RLCR and FQ-UCR — which apply Q-Learning
915 *within* the clustering paradigm — achieve lower steady-state control overhead.
916 Neither approach dominates the other across all metrics; each makes a coherent
917 trade.

918 This contradicts the intuition that clustering is uniformly superior for
919 lifetime. Clustering concentrates forwarding at a small set of cluster heads,
920 which must then be rotated to avoid premature death; under a realistic lossy
921 channel this concentration produces the rapid post-FND decay visible in
922 RLCR and FQ-UCR (HND of 1305 and 1172, the two lowest of all protocols).
923 CARL-WSN instead distributes forwarding across many energy-weighted
924 relay paths and defers traffic only at near-depleted nodes, which delays both
925 the first death and the progression to half-death. The cost of this flat-routing
926 approach is higher per-packet control overhead than clustering’s amortised
927 steady state — but that overhead remains far below the flooding- and table-
928 based protocols, and it buys a lifetime advantage that clustering does not
929 provide.

930 The broader implication for RL-based WSN routing is therefore not that
931 paradigm switching is universally more efficient, but that the choice of *what*
932 the agent controls — the routing paradigm itself, versus decisions within a
933 fixed paradigm — determines which region of the cost–lifetime trade space
934 the protocol occupies. CARL-WSN’s contribution is to show that learned

935 per-packet paradigm selection on a flat topology can extend lifetime beyond
 936 clustering while keeping overhead well below classical flat protocols.

937 *7.2. Energy-Aware Relay Selection and Network Lifetime*

938 CARL-WSN’s energy-aware relay selection in (6) addresses a well-known
 939 vulnerability of flat-routing protocols: nodes near the base station forward
 940 disproportionate traffic and deplete early. The classical flat protocols (AODV,
 941 DSDV, ZRP) share an identical FND of 327 under mixed traffic, since they
 942 select next hops on geographic progress alone without regard to residual
 943 energy.

944 By weighting relay selection toward residual energy as well as progress,
 945 and by bounding deferral at near-depleted nodes, CARL-WSN raises FND
 946 to 418 — the highest of all eight protocols, ahead of the energy-harvesting
 947 (EH-Routing, 380) and clustering (RLCR, 390) baselines, and 28% above the
 948 classical flat protocols. On half-node-death, CARL-WSN reaches HND = 2198,
 949 second only to MSLBA (2340) and well above the classical protocols (1877)
 950 and both clustering protocols (1305 and 1172). The combination of energy-
 951 aware relaying and bounded deferral thus delays both the onset and the
 952 progression of network failure, giving a flat-routing protocol a lifetime profile
 953 competitive with, and on FND exceeding, the dedicated lifetime-oriented
 954 baselines — while retaining the low control overhead characteristic of flat
 955 routing relative to flooding and table maintenance.

956 It is important to state the cost of this lifetime gain plainly. The bounded
 957 deferral mechanism withholds a fraction of non-urgent Class B/C packets at
 958 near-depleted nodes, which reduces packet delivery to 90.2% under mixed
 959 traffic, roughly five points below the baseline group. The lifetime advantage
 960 and the delivery reduction are two sides of the same mechanism; the operating
 961 point adopted here was chosen (Section 6.10) to keep the delivery cost modest
 962 while preserving the lifetime gain.

963 *7.3. Safety Constraints in RL-Based Routing*

964 A practical concern with deploying RL agents in safety-critical systems is
 965 the risk that the learned policy may violate hard constraints during exploration
 966 or after convergence to a suboptimal policy. CARL-WSN addresses this
 967 through the hard safety constraint in (20): Class A emergency packets are
 968 routed via proactive mode unconditionally, bypassing the ε -greedy selection
 969 entirely. This constraint is not a limitation of the RL agent but a deliberate

970 architectural decision that separates safety-critical routing (rule-based) from
 971 efficiency-optimised routing (learned).

972 Neither RLCR [8] nor FQ-UCR [9] implements such a constraint because
 973 neither protocol differentiates traffic by urgency class. In a deployment where
 974 both emergency and routine traffic coexist, the absence of safety constraints
 975 means that the RL agent’s exploration phase could route emergency packets
 976 via suboptimal paths with unacceptable latency. CARL-WSN eliminates this
 977 risk by design.

978 *7.4. Q-Learning Convergence Behaviour*

979 The convergence analysis in Fig. 7 shows that the Q-table stabilises within
 980 approximately 1150 rounds, with maximum update magnitudes decreasing
 981 from 10^0 to 10^{-3} . The relatively fast convergence is attributable to the
 982 compact state space (18 states): with only 54 Q-values to learn, the agent
 983 visits each state-action pair frequently enough to achieve convergence early
 984 in the network’s lifetime, well before half-node death and while most nodes
 985 remain alive.

986 The epsilon decay schedule ($\varepsilon_0 = 0.1$, $\delta = 0.998$, $\varepsilon_{\min} = 0.01$) balances
 987 exploration and exploitation effectively. After 1150 rounds, ε has reached
 988 its floor $\varepsilon_{\min} = 0.01$, meaning the agent selects the learned best action
 989 approximately 99% of the time while still occasionally exploring alternatives.
 990 This residual exploration allows the agent to adapt if network conditions
 991 change — for example, if relay nodes die and previously optimal paths become
 992 unavailable. The learned policy in Fig. 8 is contextually meaningful, selecting
 993 proactive routing for emergency traffic, reactive routing for non-urgent traffic
 994 on moderate-to-healthy nodes, and hybrid routing as an intermediate choice
 995 for moderate-energy telemetry nodes. These preferences emerge entirely from
 996 the reward signal without any manually programmed rules — the agent
 997 independently discovers that different traffic classes and network conditions
 998 warrant different routing paradigms.

999 *7.5. Context-Awareness as a Unique Capability*

1000 Table 5 demonstrates that CARL-WSN is the only protocol whose overhead
 1001 varies with traffic class composition. This context-awareness has practical
 1002 implications beyond overhead reduction. In a real smart city deployment,
 1003 traffic composition changes dynamically: a fire event increases Class A traffic
 1004 temporarily, while normal conditions are dominated by Class B telemetry.
 1005 CARL-WSN automatically adjusts its routing behaviour to match the current

traffic mix — using more proactive routing during emergencies and switching to energy-efficient reactive routing during normal operations. This adaptive overhead management is a direct consequence of the class-dependent reward function and cannot be replicated by protocols that treat all traffic identically.

7.6. Practical Deployment Considerations

CARL-WSN requires each node to maintain a Q-table of 216 bytes, an ACK history buffer of 80 bytes, and state computation variables of 12 bytes — a total memory overhead of 308 bytes beyond the standard routing table. This is deployable on widely available sensor platforms including TelosB (10 KB RAM), MicaZ (4 KB RAM), and Arduino-class devices without modification.

The Q-Learning agent operates in $O(1)$ time per packet, adding under 0.1 ms to the per-packet processing time. In a network generating one packet per node per round, the total agent computation across 100 nodes is under 10 ms per round — negligible compared to the radio transmission time.

Regarding scalability, CARL-WSN’s context classifier operates in $O(1)$ time regardless of network size N , as the state is computed from local information only (the node’s own energy, its own link stability, and the packet’s class tag). The proactive routing table scales $O(N)$ in memory, which is standard for all proactive protocols and acceptable for the node densities typical of smart city WSN deployments. The Q-table size (18×3) is independent of N — adding more nodes does not increase the state space. CARL-WSN’s overhead advantage over the flooding- and table-based protocols is expected to persist or grow with larger N : AODV’s flood cost scales with network size, whereas CARL-WSN’s per-packet control depends only on the local neighbour count. Relative to the clustering protocols, whose amortised overhead is lower than CARL-WSN’s, the comparison at larger N is an open question that the present single-size study cannot settle, and which we identify as future work.

7.7. Limitations

This study has five limitations that should be acknowledged. First, while all protocols are evaluated on a common distance-dependent lossy-channel model with per-hop acknowledgement and retransmission — a substantial step beyond idealised delivery — this model does not capture multipath fading, shadowing, temporal correlation, or interference from co-deployed networks. Second, node mobility is not modelled — all nodes are static after deployment; mobile WSN topologies where link stability varies dynamically

due to node movement represent a natural extension. Third, the latency values are simulated rather than measured on physical hardware. Fourth, all results are reported for a single network size ($N = 100$); scalability to larger deployments ($N = 500+$) is left for future validation. Fifth, the Q-Learning agent uses a fixed reward weight vector per traffic class; an adaptive weight-tuning mechanism that adjusts weights based on network congestion or remaining energy budget could further improve performance.

Future work should validate CARL-WSN on a real sensor testbed (e.g., Raspberry Pi or Arduino with IEEE 802.15.4 radios) to confirm that the simulated performance advantages hold under realistic channel conditions and hardware constraints.

8. Conclusion

This paper presented CARL-WSN, a Context-Aware Reinforcement Learning protocol for hybrid routing in heterogeneous IoT-driven Wireless Sensor Networks. CARL-WSN formulates routing paradigm selection as a Markov Decision Process and uses a tabular Q-Learning agent to dynamically select between proactive, reactive, and hybrid routing on a per-packet basis. The agent observes a three-dimensional context vector comprising application urgency class, normalised residual energy, and link stability score, and learns class-dependent routing policies through a multi-objective reward function that weights latency, overhead, and energy preservation according to traffic priority. A hard safety constraint guarantees sub-100 ms delivery for emergency traffic regardless of the learned policy, and energy-aware relay selection combined with bounded critical-node packet deferral extends network lifetime beyond all evaluated baselines.

A distinguishing feature of the evaluation is that all eight protocols are re-implemented on a common distance-dependent lossy-channel model with per-hop acknowledgement and bounded retransmission, so that packet delivery, latency, and energy consumption emerge from identical physical assumptions rather than from protocol-specific idealisations. Across four traffic scenarios and 20 independent seeds (640 runs), CARL-WSN’s behaviour under this unified methodology can be summarised as follows:

- **Longest pre-failure lifetime.** CARL-WSN attains the highest mean first-node-death round of all eight protocols ($FND = 418$), 28% above the classical flat protocols, and the second-highest mean half-node-death round ($HND = 2198$), behind only MSLBA. Tukey HSD analysis

1078 confirms these lifetime gains are statistically significant over the classical
1079 protocols and FQ-UCR (and, for HND, over RLCR), while CARL-WSN
1080 remains statistically comparable to the energy-oriented baselines EH-
1081 Routing and MSLBA.

1082 • **Low emergency latency.** CARL-WSN delivers Class A traffic in
1083 24.2 ms, a 25.6% reduction versus AODV and among the lowest of all
1084 protocols, with the safety constraint guaranteeing sub-100 ms delivery
1085 unconditionally.

1086 • **Low overhead relative to flooding and table maintenance.**
1087 CARL-WSN incurs 0.53 control packets per data packet, far below
1088 AODV (14.82), ZRP (5.91), and DSDV (2.31). It does not match the
1089 clustering protocols RLCR and FQ-UCR, whose control traffic is amor-
1090 tised over long re-clustering intervals (0.04–0.18 per packet); this is an
1091 inherent benefit of the clustering paradigm that CARL-WSN trades for
1092 per-packet adaptation and freedom from cluster-head energy hotspots.

1093 • **Context-adaptive overhead.** CARL-WSN is the only evaluated
1094 protocol whose overhead varies with traffic-class composition — from
1095 1.00 for pure emergency traffic to 0.32 for pure non-urgent traffic — em-
1096 pirically confirming that the agent learns class-specific routing policies.

1097 • **A modest, bounded delivery trade.** These gains are obtained
1098 at a packet-delivery ratio of 90.2% under mixed traffic, roughly five
1099 points below the baseline group, arising from the deferral of non-urgent
1100 traffic at near-depleted nodes. On the neutral metrics, CARL-WSN is
1101 statistically comparable to all baselines in energy consumption, shows
1102 no significant energy-balance difference, and matches the classical and
1103 energy-harvesting protocols in throughput (below MSLBA, above the
1104 clustering protocols).

1105 The Q-table converges within approximately 1150 rounds, occupies only
1106 216 bytes of RAM in single precision, and the agent operates in $O(1)$ time
1107 per packet, making CARL-WSN deployable on resource-constrained sensor
1108 hardware without modification. The learned policy autonomously discovers
1109 that emergency traffic is best served by proactive routing, non-urgent traffic
1110 by reactive routing, and moderate-energy telemetry nodes by an intermediate
1111 hybrid strategy — all without manually programmed rules.

The central finding of this work is that the choice of *what* a reinforcement-learning agent controls — the routing paradigm itself, rather than decisions within a fixed paradigm — determines which region of the cost–lifetime trade space a protocol occupies. By learning per-packet paradigm selection on a flat topology, CARL-WSN extends network lifetime beyond clustering-based protocols and protects emergency-traffic latency, while keeping control overhead well below classical flooding- and table-based protocols, at the cost of a small reduction in best-effort delivery. This positions paradigm switching not as a universally dominant strategy but as a distinct and well-characterised operating point for heterogeneous, latency- and lifetime-critical IoT-WSN deployments.

Four directions for future work are identified. First, validating CARL-WSN on a physical sensor testbed to confirm that the simulated advantages hold under real channel and hardware conditions. Second, extending the protocol to mobile WSN topologies where link stability varies dynamically due to node movement. Third, scaling the evaluation to larger networks ($N = 500+$), where the overhead comparison against clustering protocols in particular remains an open question. Fourth, investigating adaptive reward-weight tuning that adjusts the class-dependent weight vectors based on real-time network congestion and remaining energy budget.

References

- [1] I. F. Akyildiz, W. Su, Y. Sankarasubramaniam, E. Cayirci, Wireless sensor networks: a survey, *Computer Networks* 38 (4) (2002) 393–422. doi:10.1016/S1389-1286(01)00302-4.
- [2] N. Zohrabi, et al., A review of IoT-based smart city development and management, *Smart Cities* 7 (3) (2024) 61. doi:10.3390/smartcities7030061.
- [3] C. E. Perkins, P. Bhagwat, Highly dynamic destination-sequenced distance-vector routing for mobile computers, in: *Proc. ACM SIGCOMM*, 1994, pp. 234–244.
- [4] S. R. Das, C. E. Perkins, E. M. Belding-Royer, Ad hoc on-demand distance vector (AODV) routing, Request for Comments RFC 3561, Internet Engineering Task Force (2003). doi:10.17487/RFC3561. URL <https://datatracker.ietf.org/doc/rfc3561>

- 1146 [5] D. B. Johnson, D. A. Maltz, Dynamic source routing in ad hoc wireless
1147 networks, in: T. Imielinski, H. F. Korth (Eds.), Mobile Computing, Vol.
1148 353 of The Kluwer International Series in Engineering and Computer
1149 Science, Kluwer Academic Publishers, Boston, MA, 1996, pp. 153–181.
1150 doi:10.1007/978-0-585-29603-6_5.
- 1151 [6] I. M. Selim, N. S. Abdelrehem, W. M. Alayed, H. M. Elbadawy,
1152 R. A. Sadek, MANET routing protocols’ performance assessment un-
1153 der dynamic network conditions, Applied Sciences 15 (6) (2025) 2891.
1154 doi:10.3390/app15062891.
1155 URL <https://www.mdpi.com/2076-3417/15/6/2891>
- 1156 [7] Z. J. Haas, M. R. Pearlman, The performance of query control schemes
1157 for the zone routing protocol, IEEE/ACM Trans. Networking 9 (4) (2001)
1158 427–438.
- 1159 [8] I. Daanoune, A. Baghdad, Intelligent energy-aware routing in IoT-based
1160 WSNs by reinforcement learning, Ad Hoc Networks 188 (2026) 104244.
1161 doi:10.1016/j.adhoc.2026.104244.
1162 URL [https://www.sciencedirect.com/science/article/pii/](https://www.sciencedirect.com/science/article/pii/S1570870526001101)
1163 [S1570870526001101](https://www.sciencedirect.com/science/article/pii/S1570870526001101)
- 1164 [9] Z. Wang, J. Duan, An unequal clustering and multi-hop routing protocol
1165 based on fuzzy logic and Q-learning in WSNs, Entropy 27 (2) (2025) 118.
1166 doi:10.3390/e27020118.
1167 URL <https://www.mdpi.com/1099-4300/27/2/118>
- 1168 [10] A. M. Khedr, O. Alfawaz, P. V. Pravija Raj, W. Osamy, Advancing
1169 IoT-driven WSNs with context-aware routing: A comprehensive review,
1170 Computer Science Review 58 (2025) 100803.
- 1171 [11] K. Akkaya, M. Younis, A survey on routing protocols for wireless sensor
1172 networks, Ad Hoc Networks 3 (3) (2005) 325–349. doi:10.1016/j.
1173 [adhoc.2003.09.010](https://doi.org/10.1016/j.adhoc.2003.09.010).
- 1174 [12] T. H. Clausen, P. Jacquet, Optimized link state routing protocol (OLSR),
1175 Request for Comments RFC 3626, Internet Engineering Task Force (2003).
1176 doi:10.17487/RFC3626.
1177 URL <https://datatracker.ietf.org/doc/rfc3626>

- 1178 [13] Nurul Hidayah Ahmad Zukri, Ahmad Suki Che Mohamed Arif, Mo-
 1179 hammed Alsamman, Comparative analysis of reactive and proactive
 1180 mobility management in named data networking, *IAENG Int. J. Com-*
 1181 *puter Science* 52 (2) (2025).
- 1182 [14] M. Karthikeyan, D. Manimegalai, K. Rajagopal, Optimizing energy
 1183 efficiency in wireless sensor networks: dynamic routing with capuchin
 1184 search algorithm, *Multimedia Tools and Applications* 84 (15) (2025)
 1185 15453–15477. doi:10.1007/s11042-024-19625-7.
 1186 URL <https://doi.org/10.1007/s11042-024-19625-7>
- 1187 [15] S. K. Chaurasiya, S. Mondal, A. Biswas, A. Nayyar, M. A. Shah, R. Baner-
 1188 jee, An energy-efficient hybrid clustering technique (EEHCT) for IoT-
 1189 based multilevel heterogeneous wireless sensor networks, *IEEE Access* 11
 1190 (2023) 25941–25958. doi:10.1109/ACCESS.2023.3254594.
 1191 URL <https://ieeexplore.ieee.org/document/10064298/>
- 1192 [16] C. Lei, An energy-aware cluster-based routing in the internet of things
 1193 using particle swarm optimization algorithm and fuzzy clustering, *J.*
 1194 *Engineering and Applied Science* 71 (1) (2024) 135. doi:10.1186/
 1195 s44147-024-00464-0.
 1196 URL <https://doi.org/10.1186/s44147-024-00464-0>
- 1197 [17] P. Nadanam, N. Kumarathan, M. Anousouya Devi, Innovative hybrid
 1198 framework for routing and clustering in wireless sensor networks with
 1199 quantum optimization, *Cybernetics and Systems* 57 (3) (2026) 449–484.
 1200 doi:10.1080/01969722.2025.2521710.
 1201 URL <https://doi.org/10.1080/01969722.2025.2521710>
- 1202 [18] C. Perera, A. Zaslavsky, P. Christen, D. Georgakopoulos, Context aware
 1203 computing for the internet of things: A survey, *IEEE Communications*
 1204 *Surveys & Tutorials* 16 (1) (2014) 414–454. doi:10.1109/SURV.2013.
 1205 042313.00197.
 1206 URL <https://ieeexplore.ieee.org/abstract/document/6512846>
- 1207 [19] A. Monteiro, E. Souto, R. Pazzi, M. Nogueira, Context-aware network
 1208 selection in heterogeneous wireless networks, *Computer Communications*
 1209 135 (2019) 1–15. doi:10.1016/j.comcom.2018.11.006.
 1210 URL [https://linkinghub.elsevier.com/retrieve/pii/](https://linkinghub.elsevier.com/retrieve/pii/S014036641730155X)
 1211 [S014036641730155X](https://linkinghub.elsevier.com/retrieve/pii/S014036641730155X)

- 1212 [20] M. M. Sithik, B. M. Kumar, Intelligent agent based virtual clus-
 1213 tering and multi-context aware routing for congestion mitigation in
 1214 secure RPL-IoT environment, *Ad Hoc Networks* 137 (2022) 102972.
 1215 doi:10.1016/j.adhoc.2022.102972.
 1216 URL [https://www.sciencedirect.com/science/article/pii/](https://www.sciencedirect.com/science/article/pii/S1570870522001470)
 1217 [S1570870522001470](https://www.sciencedirect.com/science/article/pii/S1570870522001470)
- 1218 [21] V. Karagiannis, P. A. Frangoudis, S. Dustdar, S. Schulte, Context-aware
 1219 routing in fog computing systems, *IEEE Trans. Cloud Computing* 11 (1)
 1220 (2023) 532–549. doi:10.1109/TCC.2021.3102996.
 1221 URL <https://ieeexplore.ieee.org/abstract/document/9508866>
- 1222 [22] A. A. Hasan, X. Fang, S. Latif, A. Iqbal, Context-aware trust prediction
 1223 for optimal routing in opportunistic IoT systems, *Sensors* 25 (12) (2025)
 1224 3672. doi:10.3390/s25123672.
 1225 URL <https://www.mdpi.com/1424-8220/25/12/3672>
- 1226 [23] G. A. Akpakwu, G. P. Hancke, A. M. Abu-Mahfouz, CACC: Context-
 1227 aware congestion control approach for lightweight CoAP/UDP-based
 1228 internet of things traffic, *Transactions on Emerging Telecommunications*
 1229 *Technologies* 31 (2) (2020) e3822. doi:10.1002/ett.3822.
 1230 URL <https://onlinelibrary.wiley.com/doi/10.1002/ett.3822>
- 1231 [24] D. Karunkuzhali, B. Meenakshi, K. Lingam, OQR-SC: An optimal QoS
 1232 aware routing technique for smart cities using IoT enabled wireless sensor
 1233 networks, *Wireless Personal Communications* 125 (4) (2022) 3575–3602.
 1234 doi:10.1007/s11277-022-09725-8.
 1235 URL <https://link.springer.com/10.1007/s11277-022-09725-8>
- 1236 [25] D. Karunkuzhali, B. Meenakshi, K. Lingam, A QoS-aware routing ap-
 1237 proach for internet of things-enabled wireless sensor networks in smart
 1238 cities, *Multimedia Tools and Applications* 84 (17) (2024) 17951–17977.
 1239 doi:10.1007/s11042-024-18125-y.
 1240 URL <https://link.springer.com/10.1007/s11042-024-18125-y>
- 1241 [26] F. H. El-Fouly, M. Kachout, Y. Alharbi, J. S. Alshudukhi, A. Alanazi,
 1242 R. A. Ramadan, Environment-aware energy efficient and reliable routing
 1243 in real-time multi-sink wireless sensor networks for smart cities applica-
 1244 tions, *Applied Sciences* 13 (1) (2023) 605. doi:10.3390/app13010605.
 1245 URL <https://www.mdpi.com/2076-3417/13/1/605>

- [27] S. A. Obaid, A. J. Salman, Load-balancing in multi-sink IoT networks, *Int. J. Scientific and Engineering Research (IJSER)* 2 (2) (2023) 58–64. doi:10.58564/IJSER.2.2.2023.71. URL <https://ijser.aliraqia.edu.iq/index.php/ijser/article/view/71>
- [28] S. Hanif, A. M. Khedr, Z. Al Aghbari, D. P. Agrawal, Opportunistically exploiting internet of things for wireless sensor network routing in smart cities, *Journal of Sensor and Actuator Networks* 7 (4) (2018) 46. doi:10.3390/jsan7040046. URL <https://www.mdpi.com/2224-2708/7/4/46>
- [29] E. Inga, J. Inga, A. Ortega, Novel approach sizing and routing of wireless sensor networks for applications in smart cities, *Sensors* 21 (14) (2021) 4692. doi:10.3390/s21144692. URL <https://www.mdpi.com/1424-8220/21/14/4692>
- [30] P. Ekler, J. Levendovszky, D. Pasztor, Energy aware IoT routing algorithms in smart city environment, *IEEE Access* 10 (2022) 87733–87744. doi:10.1109/ACCESS.2022.3199757. URL <https://ieeexplore.ieee.org/document/9861603/>
- [31] R. Priyadarshi, Energy-efficient routing in wireless sensor networks: A meta-heuristic and artificial intelligence-based approach: A comprehensive review, *Archives of Computational Methods in Engineering* 31 (4) (2024) 2109–2137. doi:10.1007/s11831-023-10039-6. URL <https://doi.org/10.1007/s11831-023-10039-6>
- [32] R. Priyadarshi, R. R. Kumar, R. Ranjan, P. V. Kumar, AI-based routing algorithms improve energy efficiency, latency, and data reliability in wireless sensor networks, *Scientific Reports* 15 (1) (2025) 22292. doi:10.1038/s41598-025-08677-w. URL <https://www.nature.com/articles/s41598-025-08677-w>
- [33] M. Y. Yilmaz, B. Ustuner, I. M. Gul, H. A. Cirpan, Sustainable communication in 5g/6g wireless sensor networks: A survey on energy-efficient collaborative routing, *ITU Journal of Wireless Communications and Cybersecurity* 2 (1) (2025) 11–26. URL <https://dergipark.org.tr/en/pub/itujwcc/article/1639314>

- [34] S. Sudha, A. Srinivasan, T. G. Devi, G. Rajkumar, Intelligent routing for 5g-enabled wireless sensor networks using hybrid learning approaches, in: 2025 IEEE 17th International Conference on Computational Intelligence and Communication Networks (CICN), 2025, pp. 141–145, ISSN: 2472-7555. doi:10.1109/CICN67655.2025.11368058. URL <https://ieeexplore.ieee.org/document/11368058>
- [35] N. R. Saadallah, S. A. Alabady, A comprehensive study on energy-efficient-based routing protocols in the internet of things part i: definition and classification, Iran Journal of Computer Science 7 (3) (2024) 631–661. doi:10.1007/s42044-024-00180-1. URL <https://doi.org/10.1007/s42044-024-00180-1>
- [36] P. Kumar, A. Sharma, R. Grover, N. Kumar, V. Kuchhal, S. Singh, Exploring energy-efficient routing in IoT-based WSNs: A WoS bibliometric-based review, in: 2025 8th International Conference on Computing Methodologies and Communication (ICCMC), 2025, pp. 401–406. doi:10.1109/ICCMC65190.2025.11140596. URL <https://ieeexplore.ieee.org/abstract/document/11140596>
- [37] A. Chu, T. Zhang, C. Wang, LEACH protocol evolution in WSN: A review of energy consumption optimization and security reinforcement, Sensors 26 (7) (2026) 2272. doi:10.3390/s26072272. URL <https://www.mdpi.com/1424-8220/26/7/2272>
- [38] P. Gupta, S. Tripathi, S. Singh, Energy-efficient routing protocols for cluster-based heterogeneous wireless sensor network (Het-WSN)—strategies and challenges: A review, in: A. Khanna, D. Gupta, Z. Pólkowski, S. Bhattacharyya, O. Castillo (Eds.), Data Analytics and Management, Vol. 54, Springer Singapore, 2021, pp. 853–878, series Title: Lecture Notes on Data Engineering and Communications Technologies. doi:10.1007/978-981-15-8335-3_65. URL http://link.springer.com/10.1007/978-981-15-8335-3_65
- [39] C. M. Prashanth, et al., Cluster based energy efficient routing protocol for heterogeneous wireless sensor networks, Concurrency and Computation: Practice and Experience 35 (7) (2023) e7693. doi:10.1002/cpe.7693.
- [40] M. M. Saleh, R. S. Abdulrahman, A. J. Salman, Energy-harvesting and energy-aware routing algorithm for heterogeneous energy WSNs,

- 1314 Indonesian J. Electrical Engineering and Computer Science (IJECS)
1315 24 (2) (2021) 910–920.
- 1316 [41] W. Heinzelman, A. Chandrakasan, H. Balakrishnan, Energy-efficient
1317 communication protocol for wireless microsensor networks, in: Proc.
1318 33rd Annual Hawaii Int. Conf. System Sciences (HICSS), IEEE, 2000,
1319 pp. 1–10.
- 1320 [42] M. Zuniga, B. Krishnamachari, Analyzing the transitional region in
1321 low power wireless links, in: 2004 First Annual IEEE Communications
1322 Society Conference on Sensor and Ad Hoc Communications and Networks
1323 (SECON), 2004, pp. 517–526. doi:10.1109/SAHCN.2004.1381954.
- 1324 [43] IEEE, IEEE standard for low-rate wireless networks, IEEE Std 802.15.4-
1325 2020 (2020). doi:10.1109/IEEESTD.2020.9144691.
- 1326 [44] A. Woo, D. Culler, Evaluation of efficient link reliability estimators
1327 for low-power wireless networks, Tech. Rep. UCB/CSD-03-1270, EECS
1328 Dept., Univ. of California, Berkeley (2003).
- 1329 [45] D. Bhatt, et al., Implementation of novel learning based energy ef-
1330 ficient routing protocols in wireless sensor networks for Internet of
1331 Things use cases, Discover Computing 28 (1) (2025) 1–24. doi:
1332 10.1007/s10791-025-09718-8.

Algorithm 1: CARL-WSN Per-Packet Routing with Q-Learning

Input: Packet p with class U , node i , network state $\mathcal{N}$, Q-table Q , exploration rate ε , per-node deferral counter c_{def}^i (init. 0)

Output: Updated network state, updated Q-table

```
// Step 1: Compute context vector
1  $U \leftarrow \text{READCLASSTAG}(p);$ 
2  $E_r^i \leftarrow E_i(t)/E_0^i;$ 
3  $L_s^i \leftarrow \sum_{k=0}^{W-1} w_k \cdot a_k^i;$ 
4  $s \leftarrow \text{STATEINDEX}(U, E_r^i, L_s^i);$ 
5 if  $U = A$  then
    // Step 2: Safety constraint -- emergency traffic
    // bypasses learning
6      $a \leftarrow \text{proactive};$ 
7 else
    /* Step 3: Bounded critical-node deferral */
8     if  $E_r^i < T_{\text{defer}}$  and  $U \in \{B, C\}$  and  $c_{\text{def}}^i < \delta_{\text{max}}$  then
9          $c_{\text{def}}^i \leftarrow c_{\text{def}}^i + 1;$ 
10        Defer packet (minimal listening energy cost);
11        return;
12    else
13         $c_{\text{def}}^i \leftarrow 0;$  // forced or normal transmit ends the
        deferral run
14    end
15 end
    // Step 4:  $\varepsilon$ -greedy action selection
16 if  $\text{rand}() < \varepsilon$  then
17      $a \leftarrow \text{random action from } \mathcal{A};$ 
18 else
19      $a \leftarrow \arg \max_{a'} Q(s, a');$ 
20 end
    // Step 5: Route with energy-aware relay
21  $j^* \leftarrow \arg \max_{j \in \mathcal{N}_i} [(0.4 \cdot \text{progress}(j) + 0.6 \cdot E_r^j) \cdot \text{PRR}(d_{ij})];$ 
22 Route packet via mode  $a$  using relay  $j^*$ ;
    // Step 6: Observe outcome and compute reward
23  $R \leftarrow w_1^U \cdot R_{\text{lat}} + w_2^U \cdot R_{\text{oh}} + w_3^U \cdot R_{\text{en}};$ 
24  $s' \leftarrow \text{STATEINDEX}(U, E_r^{i, \text{new}}, L_s^{i, \text{new}});$ 
    // Step 7: Q-table update (skip for Class A)
25 if  $U \neq A$  then
26      $Q(s, a) \leftarrow Q(s, a) + \alpha [R + \gamma \max_{a'} Q(s', a') - Q(s, a)];$ 
27 end
28  $\varepsilon \leftarrow \max(\varepsilon \cdot \delta, \varepsilon_{\text{min}});$ 
```

Table 3: Simulation Parameters

Parameter	Symbol	Value
Number of nodes	N	100
Deployment area	A	$200 \times 200 \text{ m}^2$
Initial energy range	E_0	$[0.5, 2.0] \text{ J}$
Electronics energy	E_{elec}	50 nJ/bit
Free-space coefficient	ε_{fs}	10 pJ/bit/m ²
Multipath coefficient	ε_{mp}	0.0013 pJ/bit/m ⁴
Packet size	L	4000 bits
PRR midpoint	β_{ch}	90 m
PRR sharpness	α_{ch}	0.18
Max retransmissions	$R_{\max}^{ARQ}$	3
Communication range	R_{tx}	100 m
ACK window size	W	30
Simulation rounds	R	8000
Independent runs	—	20 (seeds 42–61)
<i>Q-Learning hyperparameters</i>		
Learning rate	α	0.1
Discount factor	γ	0.9
Initial exploration	ε_0	0.1
Exploration decay	δ	0.998
Minimum exploration	$\varepsilon_{\min}$	0.01
Deferral threshold	T_{defer}	0.05
Max consecutive deferrals	$\delta_{\max}$	3
EWMA smoothing factor	α_{ewma}	0.5
Relay progress weight	w_p	0.4
Relay energy weight	w_e	0.6

Table 4: Complete Results — Mixed Traffic Scenario (mean across 20 seeds)

Protocol	FND	HND	PDR	LatA (ms)	AvgE (J)	Gini	Overhead	Tput (pkt/rnd)
CARL-WSN	418	2198	90.2%	24.2	0.024	0.200	0.53	37.8
AODV	327	1877	94.9%	32.5	0.024	0.196	14.82	36.7
DSDV	327	1877	94.9%	24.4	0.024	0.196	2.31	36.7
ZRP	327	1877	94.9%	28.2	0.024	0.196	5.91	36.7
EH-Routing	380	2126	96.0%	24.4	0.024	0.203	1.00	38.2
MSLBA	362	2340	96.9%	21.1	0.024	0.193	1.00	48.5
RLCR	390	1305	94.1%	34.0	0.024	0.200	0.04[‡]	23.2
FQ-UCR	123	1172	91.5%	35.2	0.024	0.198	0.04[‡]	19.5

Bold = best value in column. [‡]Re-clustering-only accounting; 0.10–0.18 under TDMA accounting (see text).

[‡]For the clustering protocols (RLCR, FQ-UCR), the overhead column reports re-clustering control packets per data packet; their per-slot TDMA scheduling overhead (0.10–0.18) is accounted separately and shown in Fig. 3.

Table 5: Routing Overhead per Traffic Scenario — Context-Awareness Validation

Protocol	Class A	Class B	Class C	Mixed	Adapts?
CARL-WSN	1.00	0.32	0.32	0.53	Yes
AODV	14.82	14.82	14.82	14.82	No
EH-Routing	1.00	1.00	1.00	1.00	No
MSLBA	1.00	1.00	1.00	1.00	No

Table 6: CARL-WSN versus RL Baselines — Mixed Traffic (CARL-WSN value vs. baseline value)

Comparison	FND	HND	LatA (ms)	Overhead	PDR
CARL-WSN	418	2198	24.2	0.53	90.2%
RLCR	390	1305	34.0	0.04–0.18	94.1%
FQ-UCR	123	1172	35.2	0.04–0.10	91.5%

CARL-WSN leads on lifetime (FND, HND) and emergency latency;
the clustering baselines lead on overhead and slightly on PDR.

Table 7: Deferral Policy Sensitivity — Mixed Traffic (20 seeds)

T_{defer}	δ_{max}	PDR (%)	FND	HND
0.00 (off)	—	95.1	381	2117
0.05	3	90.2	418	2198
0.05	5	88.9	428	2220
0.10	3	85.1	455	2295
0.10	5	82.7	475	2342
0.15	5	76.5	525	2478
0.20	10	66.4	635	2787

Table 8: Model-Parameter Sensitivity — Metric Swing (%) Across Swept Range, Mixed Traffic, 20 Seeds

Parameter (range)	PDR	FND	HND	LatA	OH
w_e/w_p (0.4/0.6–0.8/0.2)	0.6	0.0	4.7	1.0	0.4
PRR β (80–100)	10.1	64.8	28.5	16.2	2.7
PRR α (0.12–0.25)	0.2	15.2	6.5	6.9	4.3
T_L (0.6–0.8)	0.0	0.0	0.0	0.0	0.0

Swing = $100 \times (\max - \min) / \text{mean}$ across the swept values.

Table 9: CARL-WSN Standing on Principal Metrics — Mixed Traffic

Metric	CARL-WSN result
Network lifetime (FND)	Highest of all eight protocols (418)
Network lifetime (HND)	Second-highest (2198), behind MSLBA
Class A latency	25.6% below AODV; among the lowest overall
Routing overhead	Far below AODV/ZRP/DSDV; above clustering
Packet delivery	90.2% (≈ 5 pts below baselines)
Energy consumption	Statistically comparable to all baselines
Energy balance (Gini)	No significant difference involving CARL-WSN
Throughput	Comparable to classical and EH-Routing; below MSLBA